

清华大学基础模型2025学术年会 分论坛六 – 具身智能

触觉驱动具身智能发展新引擎
方斌 北京邮电大学教授

2025年5月29日

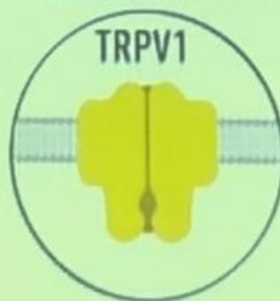


2021年诺贝尔生理学或医学奖得主
加州大学旧金山分校教授David Julius



2021年诺贝尔生理学或医学奖得主

温度感受器TRPV1



Temperature
Heat pain

温度/热疼痛

核心体温 Core body temperature
炎症疼痛 Inflammatory pain
神经性疼痛 Neuropathic pain
内脏疼痛 Visceral pain
保护性反应 Protective reflexes

压力感受器PIEZO2



Touch
Proprioception

触摸本体感觉

Mechanical pain 机械疼痛
Urination 排尿
Respiration 呼吸
Blood pressure 血压
Skeletal remodeling 骨骼重塑



在触觉、本体感觉和听觉中，某种蛋白质可能既用作力传感器，又充当离子通道。三个Piezo蛋白形成三聚体穿过细胞膜。以中央孔为中心，三个蛋白盘旋向外，就像螺旋桨叶片那样。它们向上向外弯曲，在细胞表面形成了一个深深的凹陷结构。

正是触觉信号的生成-传递-处理构建人类的触觉基础



Lateral motion:
texture



Pressure:
hardness



Static contact:
temperature



Unsupported
holding: weight



Enclosure: global
shape, volume

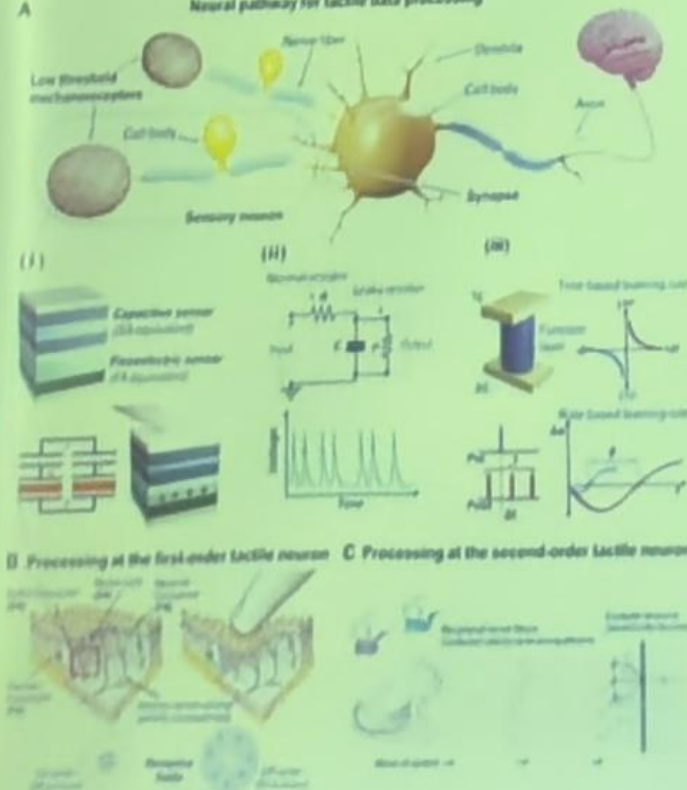


Contour following: global
shape, exact shape

进而引导人类操作行为的规划

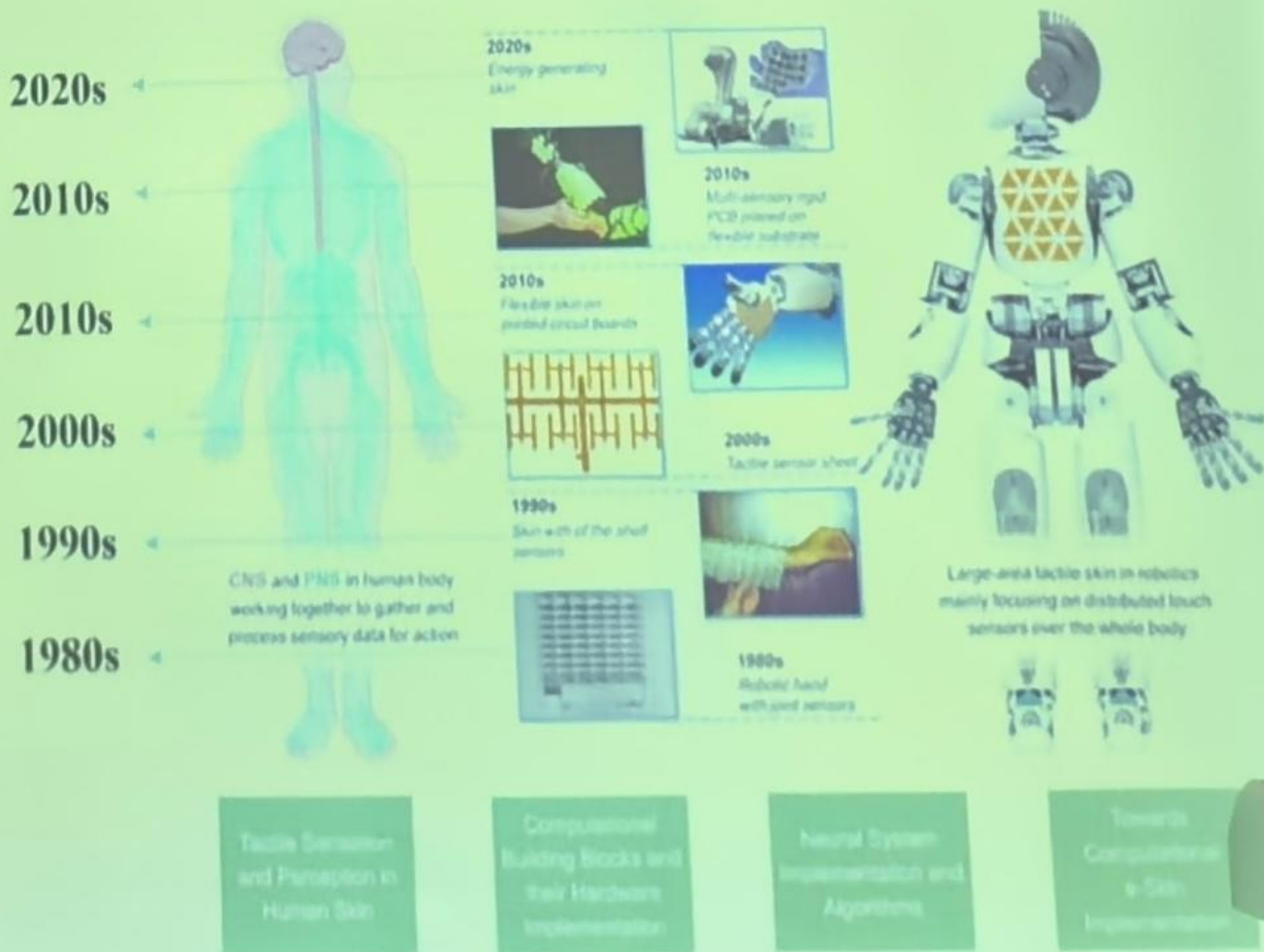
Fengyuan Liu^{1†}, Sweetey Deswal^{1†}, Adamos Christou¹, Yulia Sandamirskaya²,
Mohsen Kaboli^{3,4}, Ravinder Dahiya^{1*}

A Neural pathway for tactile data processing



人体皮肤触觉数据处理的神经通路及其使用基本器件和电路的简单实现方案。

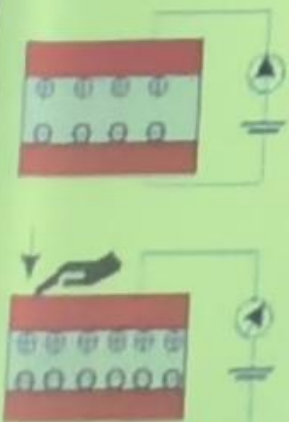
触觉与操作的紧耦合性仍存在一定的差距



触觉传感器发展现状

机器人的触觉感知

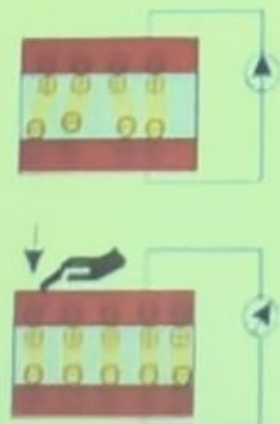
压电式



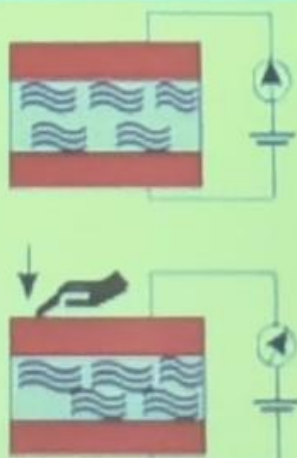
电阻应变式



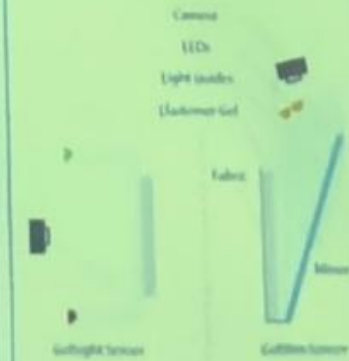
电容式



电感式



视觉式



磁触觉感知



性能指标

稳定与可靠

灵敏与量程

动态与迟滞

机器人触觉感知问题

触觉测量类型多样

电容式
压电式
光学式
压阻式

问题

如何通过多学科合作形成稳定可靠触觉传感器?

2014-2018国家自然科学基金的仪器专项

2018-2020国家科技部智能机器人重点研发计划

解决出路

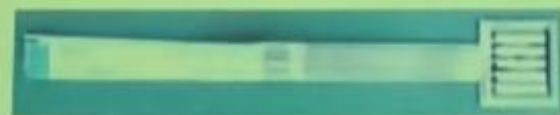
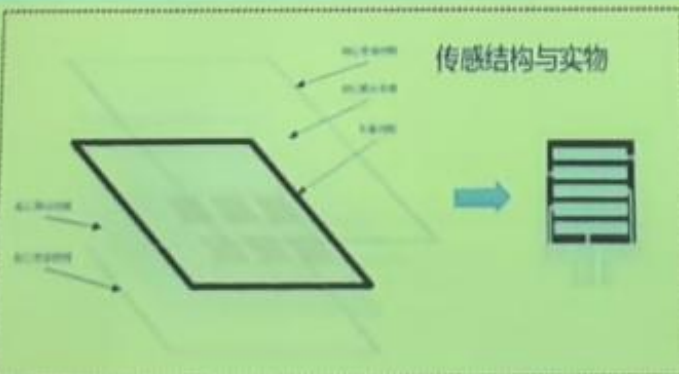
材料

电路

软件

计算

传感结构与实物



从上层到下层依次为顶层绝缘材料、顶层横向电极、压敏材料、底层纵向电极、底层绝缘材料。电极为“五横五纵”形式，从而将整块压敏材料自动分割成 5×5 矩阵形式。

电路原理与实物



阵列为公用行列线结构，数据采集电路需要对行列线进行循环选通，完成对传感单元信号的采集。



柔性触觉阵列传感器

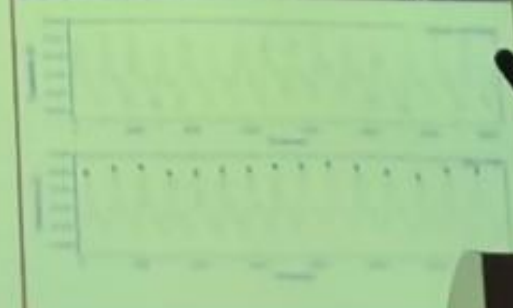
针对目前机器人触觉传感器的阵列密度低和不耐用等缺点，设计了花瓣式阵列分布结构和微针传导结构，研制的柔性触觉传感器单元密度高、集成度高。



微针传导结构



四代柔性触觉阵列



中医搭脉机理研究

- A flexible capacitive tactile sensor array with micro structure for robotic application. Science China Information Sciences, 2015, 57 (12): 1-6.
- A novel wearable tactile sensor array designed for fingertip motion recognition. ROBIO 2017: 165-170
- 发明专利，专利号：201610064011.5，201710382607.4，201410500410.2

机器人触觉感知问题

触觉大面积感知

点
接
触

线
接
触

面
接
触

问题

如何实现仿生结构的
大面积柔性触觉感知？

手掌的大面积磁触觉感知及触觉识别

解决出路

磁
触
觉
传
感
器

磁触觉传感器整体制作流程

导电颗粒

柔性硅膜

混合搅拌

含有磁颗粒的硅胶材料

倒入模具+真空排气泡

硅胶固化成型

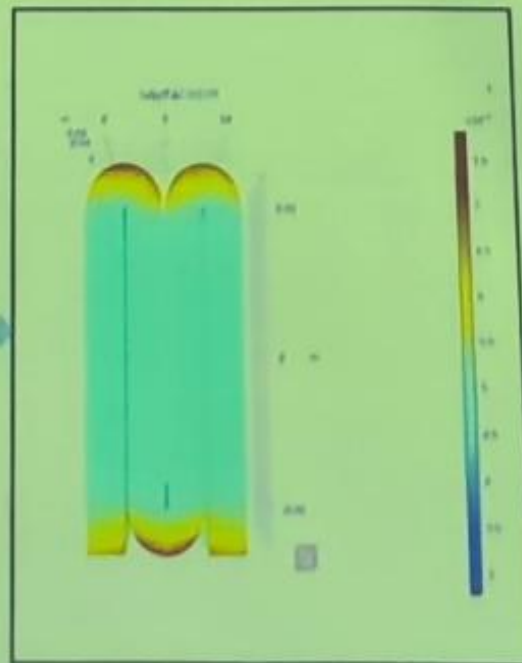
进行充磁处理

具有磁性的柔性硅胶

与传感器组装+二次成型

定制的磁触觉传感器

磁触觉手掌结构设计

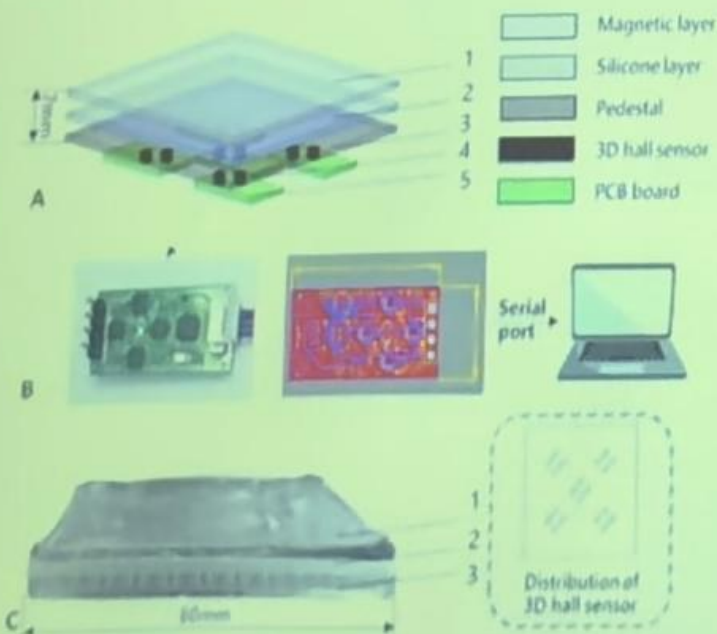


磁场分布优化及感知算法



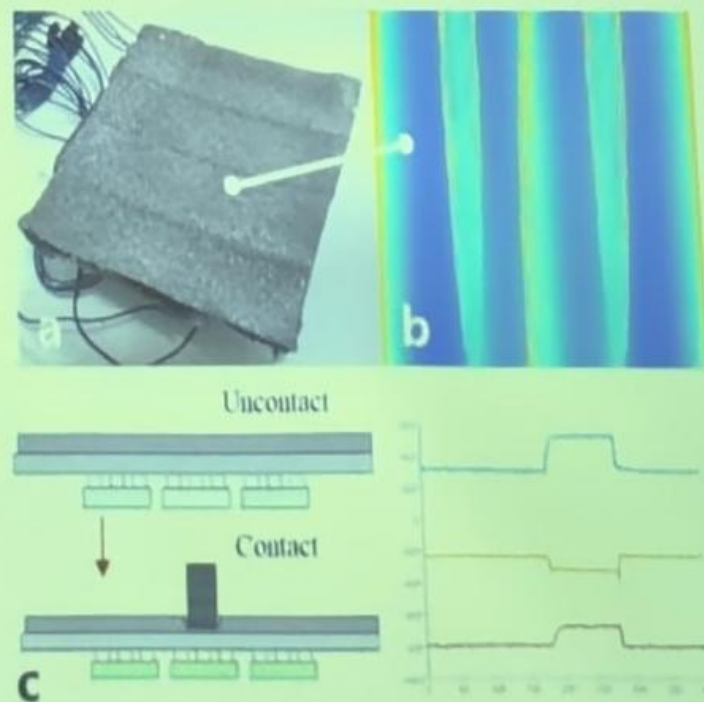
一体化集成测试

磁触觉

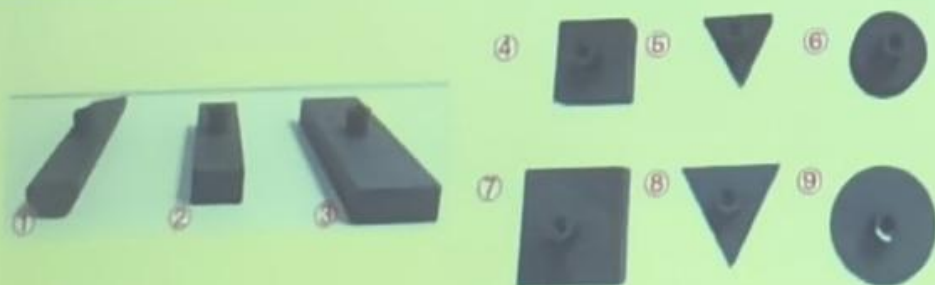


主要结构:

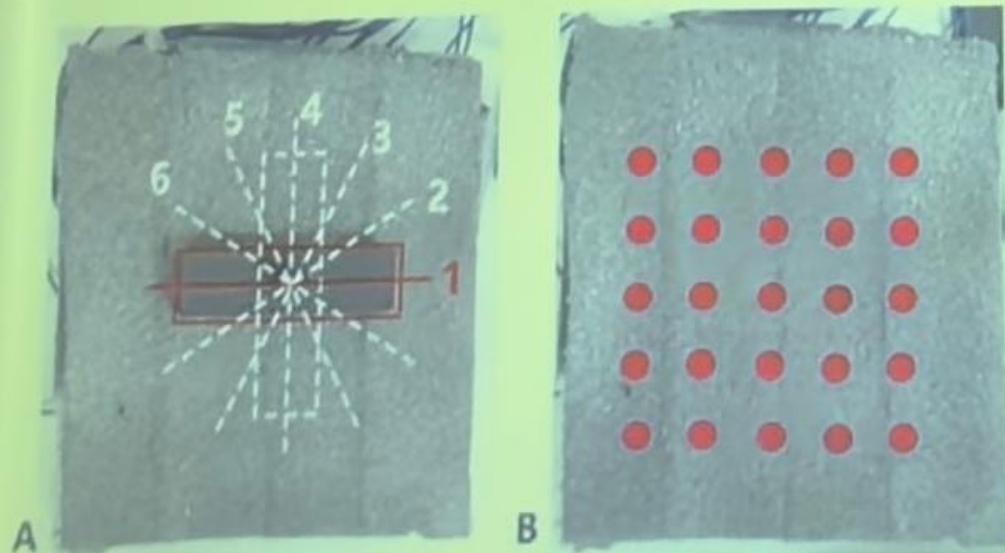
- 霍尔传感器: 检测磁信号的变化
- 磁性层: 用于检测接触信号
- 支撑层: 用于霍尔传感器的固定



- 传感器的磁场分布: 多磁极磁场
- 响应速度较快: 可以及时响应接触信号



6-ways accuracy(%)	1-shot	3-shots	5-shots
Measuring head 1 and 2 $\rightarrow$ 3	91.25	94.64	97.88
Measuring head 1 and 3 $\rightarrow$ 2	91.86	95.12	98.14
Measuring head 2 and 3 $\rightarrow$ 1	91.78	94.76	98.28



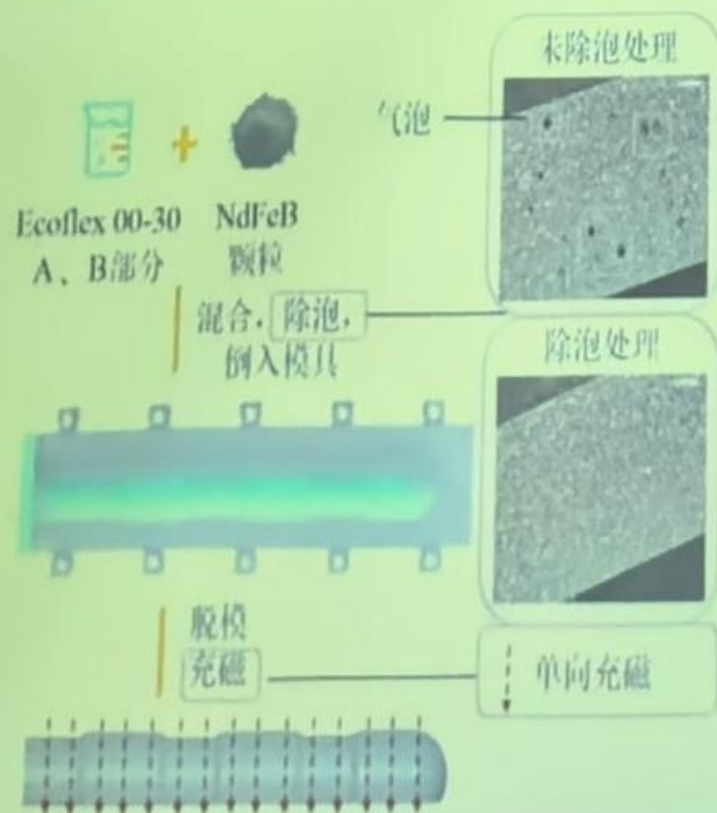
accuracy(%)	Measuring head 1 and 2 $\rightarrow$ 3	Measuring head 1 and 3 $\rightarrow$ 2	Measuring head 2 and 3 $\rightarrow$ 1
Promet(1-shot)	91.25	91.86	91.78
Promet(3-shot)	94.64	95.12	94.76
Promet(5-shot)	97.88	98.14	98.28
KNN (K-NearestNeighbor)	92.85	93.54	93.07
CNN	89.15	87.58	90.47

采集了9类触头的接触信息：包括接触位置、接触姿态两部分内容

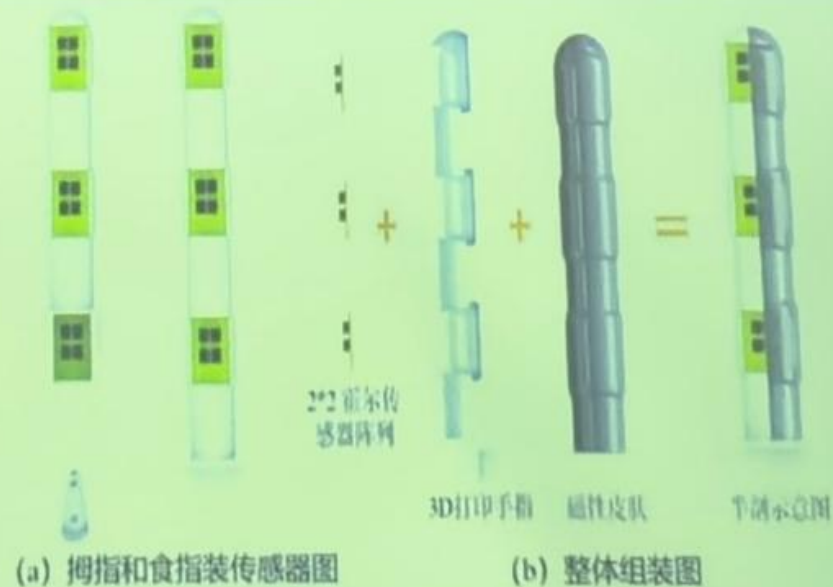
测试结果：能较为准确的完成不同触头的接触位置识别任务。在姿态、形状两个识别任务上表现良好。

Z. Xia, B. Fang, et al., "Contact Shape and Pose Recognition: Utilizing a Multipole Magnetic Tactile Sensor with a Metalearning" in IEEE Robotics & Automation Magazine, doi: 10.1109/MRA.2022.3198372. (入选期刊前50最受关注论文)

原理：当皮肤与物体发生接触或产生变形时，内部磁性颗粒发生位移重组，磁场分布发生改变，三维霍尔传感器可以检测出磁场的变化。

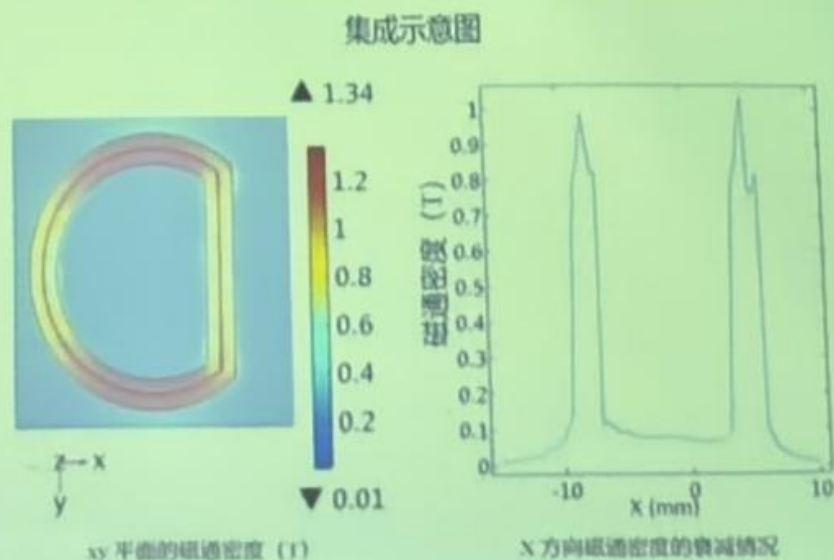


制备软磁皮肤流程图



(a) 拇指和食指装传感器图

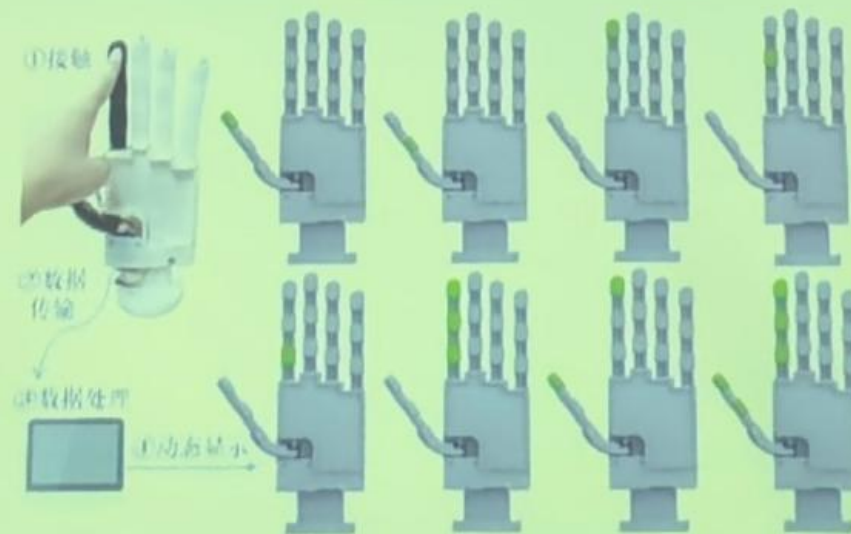
(b) 整体组装图



Consol仿真结果图



灵巧手和人手的手势图



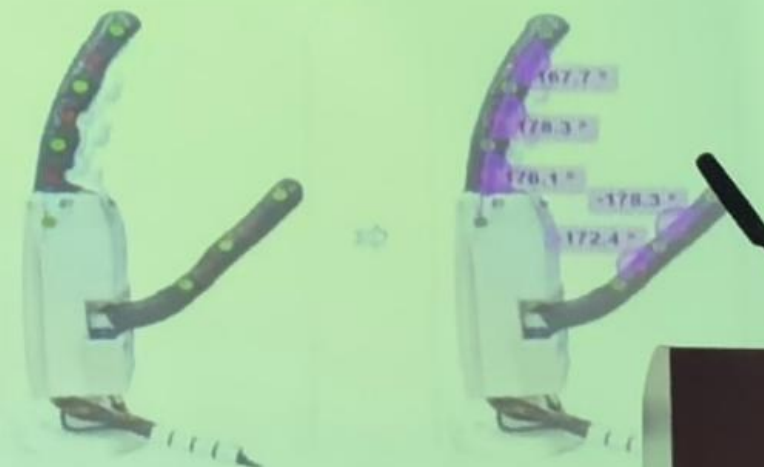
动态显示接触部位图



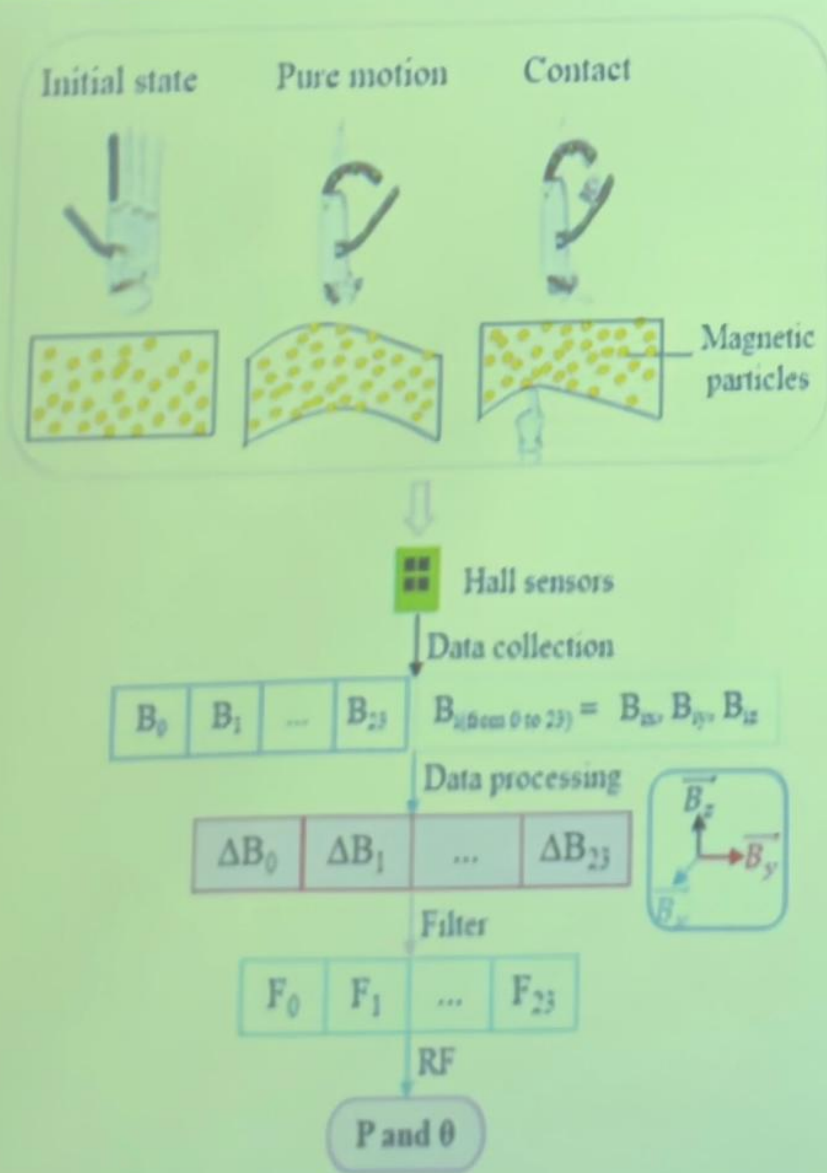
动态显示接触部位实验效果图

接触觉感知

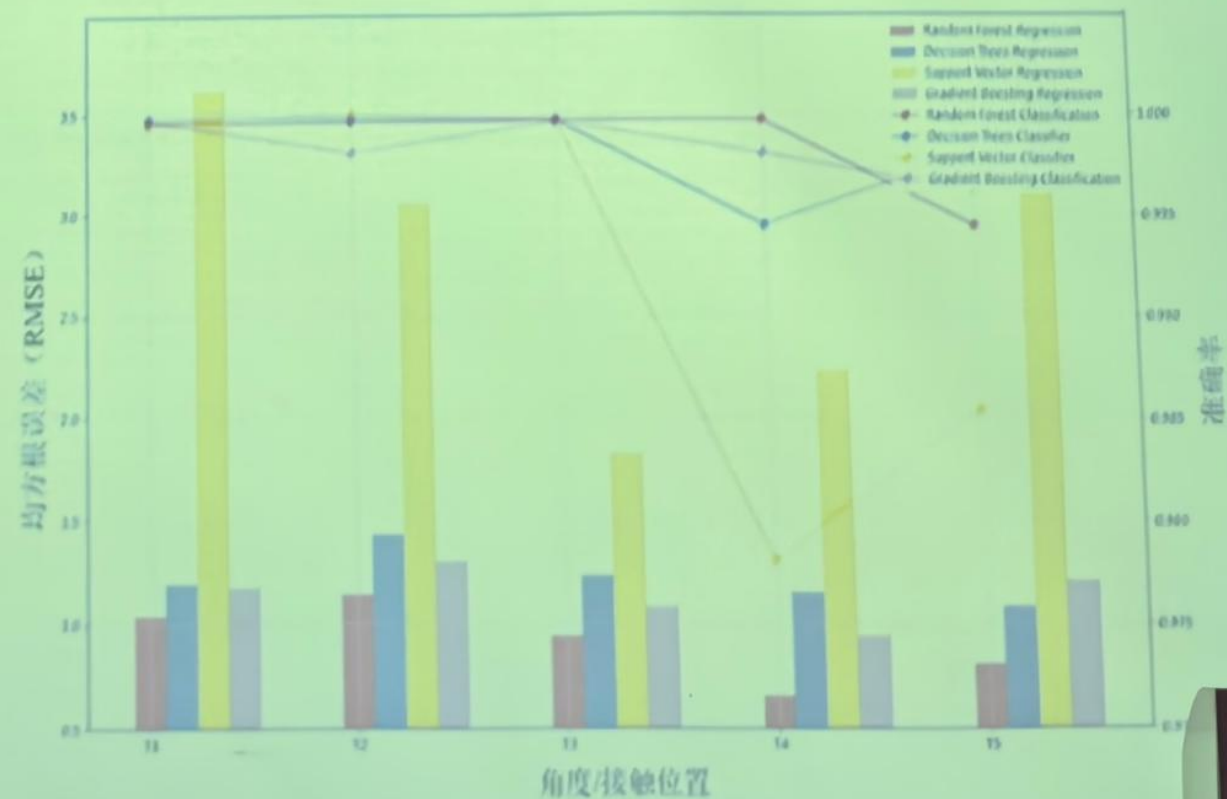
运动觉感知



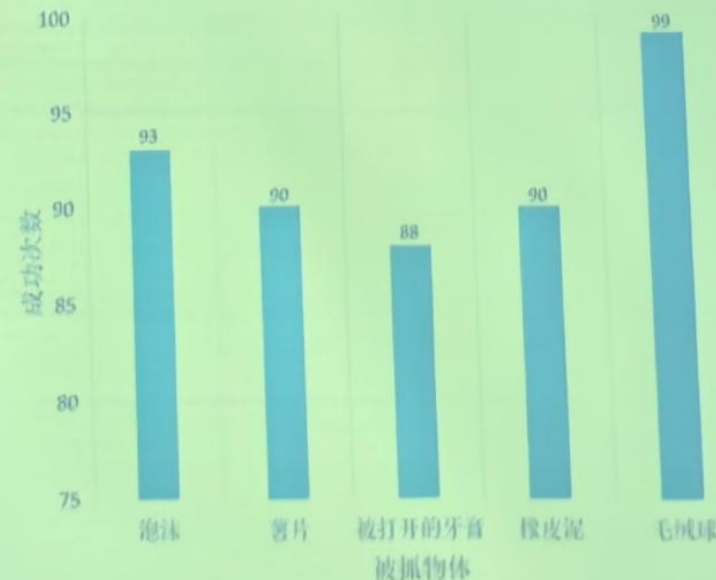
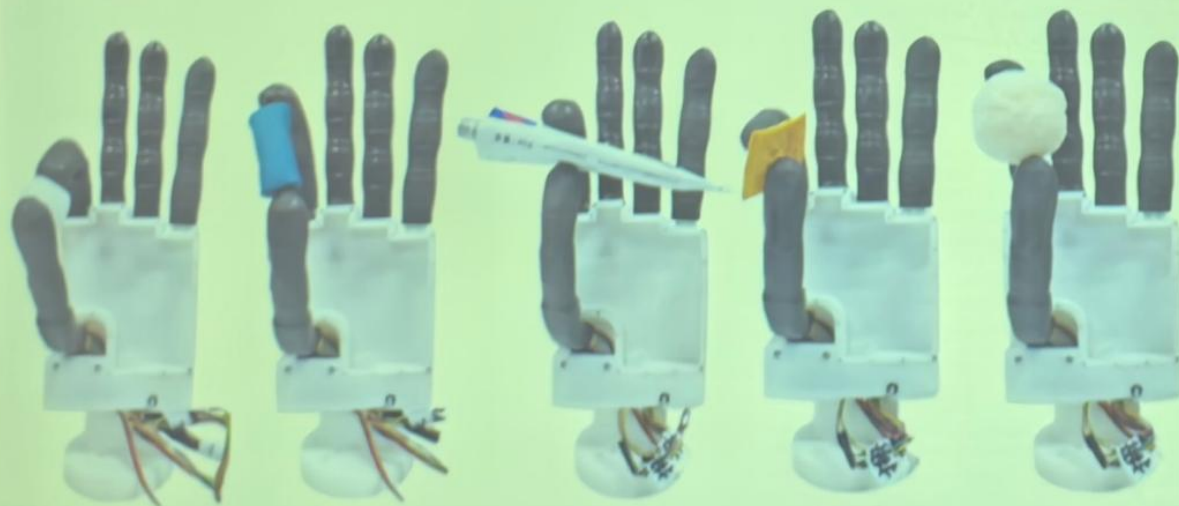
Kinovea标记关节角



对于接触位置分类的正确率最低优于99.4%；对于关节角的拟合， $RMSE = 0.9 \pm 0.2^\circ$ 。



灵巧手表面的皮肤具有柔软、弹性和可变形的特性，能够更好的适应不同形状物体表面。用相同的按压力接触物体时，软磁皮肤能减小损坏物体的风险，为灵巧手和物体提供一定的保护和缓冲效果。



我们对上述五种物体进行抓取操作，每种物体进行100次，以在抓住后轻微摇晃10 s，保持稳定抓握不掉落且平稳放下后无破损、明显或不可逆形变为标准，测试抓取成功率。测试结果如图5.10，对**泡沫、橡皮泥、打开的牙膏、薯片、毛绒球**的抓取成功分别为**93%、90%、88%、90%和99%**。

X. Ding, B. Fang^{*} et al., "Soft Magnetic Skin With Motion and Contact Sensing for Anthropomorphic Robotic Finger," *IEEE Robot Automation Letters*, vol. 9, no. 12, pp. 11601-11608, Dec. 2024, doi: 10.1109/LRA.2024.3495522

机器人触觉感知问题

触觉通道形式多样

力式触觉	纹理触觉	温度触觉
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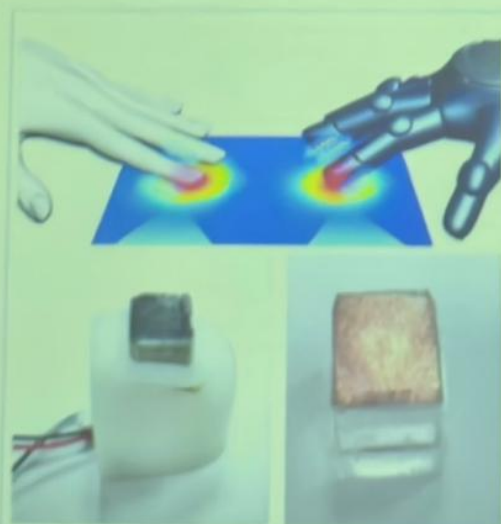
问题

如何实现高质量多模态触觉的测量与感知?

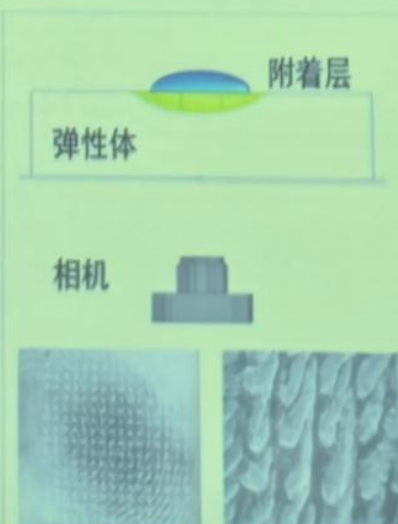
提出了图像表征弹性形变的触觉感知方法, 解决了机器人高分辨率操作触觉感知难题, 首次实现了力、纹理、温度、手内姿态等多模态高分辨率的触觉感知。

解决出路

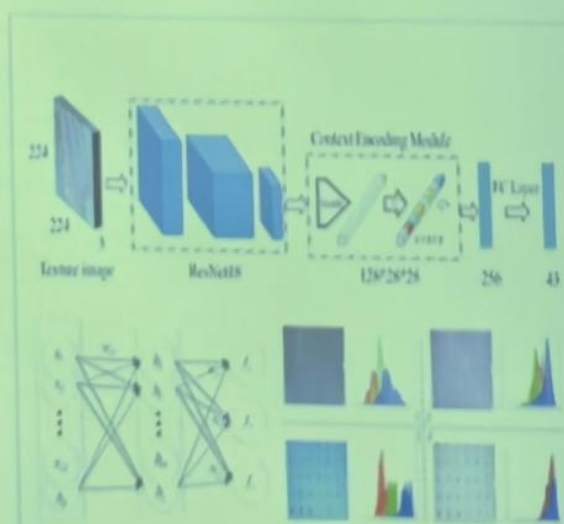
视触觉传感器



拟人接触变形
物理表征



多模态触觉
图像显现



图像分层解析的多模态触觉
感知算法



2014-2018
金属溅射



2019
热致变层



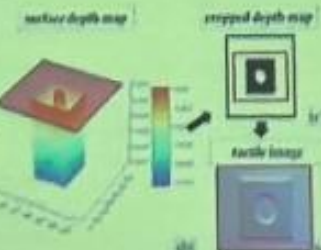
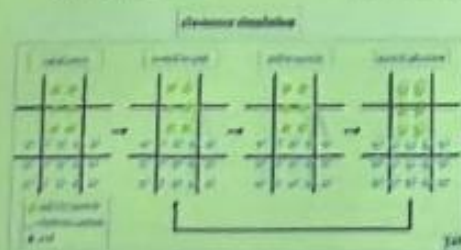
2020
内嵌软体手

2021
PaLmTac

2021
玻璃杯属性识别



2022 布料缺陷检测

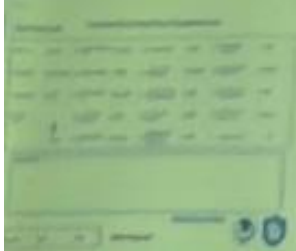
[illegible]

2023
TIRgel

2023
视触觉仿真: Tacchi 1.0

2023
贴金工艺

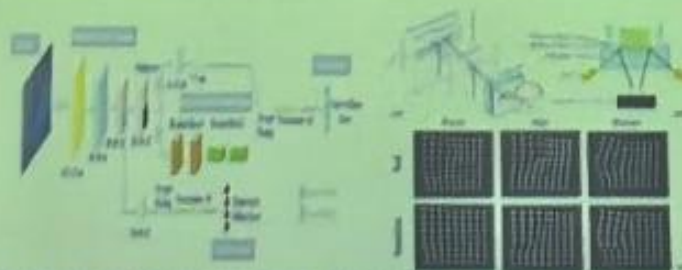
2023 视触觉力/纹理各类感知综述



2024



2024



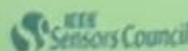
2024



2025



视触觉传感器硬件综述



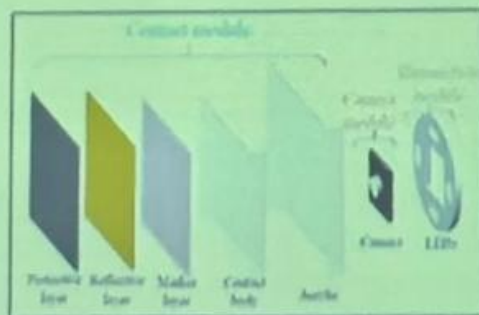
IEEE SENSORS JOURNAL, VOL. XX, NO. XX, XXXX 2017

Hardware Technology of Vision-Based Tactile Sensor: A Review

Shixin Zhang, Zixi Chen, Yuan Gao, WeiWei Wan, Senior Member, IEEE, Jianhua Shan, Hongxiang Xue, Fuchun Sun, Fellow, IEEE, Yiyong Yang, and Bin Fang, Member, IEEE,

Abstract—A vision-based tactile sensor (VBTS) is an innovative optical sensor widely applied in robotic perception. Although there has been a review of the structure design and perception methods of VBTS, the lack of details on hardware technologies cannot formulate a complete development process. Considering the rapid development of VBTSs, we reviewed the literature in the past five years and focused on hardware technologies. The VBTS consists of a contact module, an illumination module, and a camera module. This paper analyses the core components of each module from functions, material selection, and preparation processes and sorts out a technical route. We also discuss the current challenges and problems of hardware technologies and propose some feasible solutions. In cross-disciplinary applications, cutting-edge hardware technologies from other fields are expected to promote the development of next-generation VBTSs. The original intention of this paper is not only to provide a development guide for beginners but also to provide a valuable reference for the research community in improving the hardware performance of VBTSs.

Index Terms—Vision-based tactile sensor; Contact module; Illumination module; Camera module



详细回顾了近5年视觉触觉传感器的硬件设计和发展。目前，视觉触觉传感器仍面临耐用性、互换性、组装性、灵敏度、多模式、光学优化、产线组合、抗干扰等八个方面的挑战。

Zhang, S., Chen, Z., Gao, Y., Wan, W., Shan, J., Xue, H., ... & Fang, B*. "Hardware Technology of Vision-Based Tactile Sensor: A Review" in *IEEE Sensors Journal*, 2022, doi: 10.1109/JSEN.2022.3210210. (入选期刊最受关注文章之一)

视触觉传感器感知综述：纹理感知

REVIEW



www.advintelsyst.com

Application of Image Sensing System in Mineral/Rock Identification: Sensing Mode and Information Process

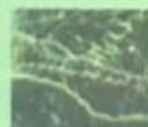
Shixin Zhang, Yiyong Yang,* Fuchun Sun, and Bin Fang*

Artificial intelligence promotes the development of mineral/rock automatic recognition technology to reduce labor costs and reliance on personal experience. With instrument digitization, image recognition has played an increasingly important role. This review focuses on the image sensing system applied to mineral/rock identification and redivides this system into a sensing module, an imaging module, and an information processing module. The sensing and imaging mechanisms are summarized via technologies of sensing, imaging, and display. The information processing is classified into single-modal or multimodal recognition. This review analyzes the limitations of single-modal recognition and discusses the advantages and significance of multimodal recognition. The challenges of current mineral/rock identification are concluded, such as public database, multimodal learning, and portable sensing system. The interdisciplinary sensing technologies are discussed, and application feasibility is explored. This review aims to provide a reference for the research community in automatic mineral/rock identification and sensing system optimization.

accurate properties of minerals/rocks, such as infrared spectroscopy, X-ray diffraction, and microscopic image analysis.^[1-5] However, recognition precision is often affected by the experience and expertise of the observer.

Artificial intelligence (AI) technology has achieved a response similar to human cognition. It has expanded into robotics, speech recognition, image recognition, natural language processing, and expert systems.^[6-7] Data-driven AI models enhance the ability of sensing systems to adapt to complex environments.^[8] In recent years, geological data have been growing geometrically.^[9] Deep learning improves the extraction of complex features. It has been applied in mineral/rock attribute

Lycopera



Trilobite



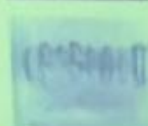
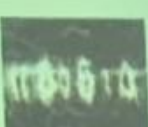
Calamite



Spiriferida



Crinoid stems



Snail



本文基于传感、成像和显示技术对传感和成像机理进行了总结，并将信息处理分为单模态识别和多模态识别。本文分析了单模态识别的局限性，讨论了多模态识别的优势和意义。本文旨在为矿物/岩石纹理自动识别和传感系统优化的研究提供参考。

视触觉传感器感知综述：力测量

REVIEW

ADVANCED
INTELLIGENT
SYSTEMS

www.advintellsyst.com

Force Measurement Technology of Vision-Based Tactile Sensor

Bin Fang,^{*} Jie Zhao, Nailong Liu, Yuhao Sun, Shixin Zhang, Fuchun Sun, Jianhua Shan, and Yiyong Yang

Marker-type vision-based tactile sensors (VTS) realize force sensing by calibrating marker vector information. The tactile visualization can provide high-precision and multimodal force information to promote robotic dexterous manipulation development. Considering VTS's contribution to force measurement, this article reviews the advanced force measurement technologies of VTSs. First, the working principle of marker-type VTSs is introduced, including single-layer markers, double-layer markers, color coding, and optical flow. Then, the relationship between the marker type and the category of force measurement is discussed in detail. On this basis, the process of marker feature extraction is summarized, including image processing and marker-matching technologies.

touch.^[5] The robot needs force sensing to obtain the interaction force between it and the external environment.^[6] For example, force feedback is used to adjust the posture and grasping force in real-time during operation. When necessary, force sensing helps the robot cope with the constraints of complex environments to improve the robot's operating precision and adaptability.^[4] Currently, electronic force sensors represented by Biotac and ATI play an important role in robot operation tasks.^[4]

详细回顾了近5年视觉触觉传感器的硬件力标定的方法。进一步讨论了传感器设计、材料和识别方法与力测量性能的相关性。最后，分析了其中的困难和挑战，以及未来的发展提出了一些建议。

Fang B^{*}, Zhao J, Liu N, et al. Force Measurement Technology of Vision-Based Tactile Sensor[J]. Advanced Intelligent Systems, 2024: 2400290.

视触觉传感器感知综述：三维重建

REVIEW

ADVANCED
FUNCTIONAL
MATERIALS
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Artificial Skin Based on Visuo-Tactile Sensing for 3D Shape Reconstruction: Material, Method, and Evaluation

Shixin Zhang, Yiyong Yang,* Yuhao Sun, Nailong Liu, Fuchun Sun, and Bin Fang*

Artificial skin has shown great potential in robot perception and human healthcare. It provides multifunctional tactile sensing, including 3D shape reconstruction, contact feedback, and temperature perception, where the 3D reconstruction function is indispensable for dexterous hands in tactile cognition and interaction. Vision-based tactile sensor (VTS) is an innovative bionic tactile sensor and supports high-resolution, high-precision, and high-density tactile reconstruction compared with electronic tactile sensors. Considering the unique contribution of visuo-tactile sensing to artificial skin, this review focuses on the 3D reconstruction techniques of the VTS. 3D reconstruction methods are classified into five categories based on sensing modalities, hardware categories, and modeling approaches: 1) photometric stereo, 2) binocular depth calibration, 3) optical flow, 4) deep learning, and 5) ToF (time of flight). In addition, the association and difference of reconstruction methods are analyzed from the hardware perspective, and the development and technological details of 3D reconstruction are summarized.

it is refracted to the retina (photosensitive tissue). Via neural system processing, these light signals are converted into electrical signals and transmitted to the brain.^[1] The primary visual cortex (area V1) further processes these signals to form a "feature map".^[2] After transmission to the cerebral cortex, the above 3D information is integrated with other sensory information (touch) to form a complete picture of the external world.^[3]

Researchers follow human perceptual patterns to build artificial sensing systems.^[4-7] Machine vision performs appearance attribute discrimination, 3D reconstruction, and object localization by simulating the stereoscopic vision of the human eye.^[8] Artificial touch takes artificial skin as a medium to explore the

详细回顾了近5年视觉触觉传感器的

硬件深度标定。目前三维重建方法

分为五类： 1)光度立体， 2)双眼深

度校准， 3) optical光流， 4) 深度

学习， 5) ToF（飞行时间）。

Zhang S, Yang Y, Sun Y, Fang B*, et al. Artificial Skin Based on Visuo-Tactile Sensing for 3D Shape Reconstruction: Material, Method and Evaluation[J]. Advanced Functional Materials, 2024: 2411686.

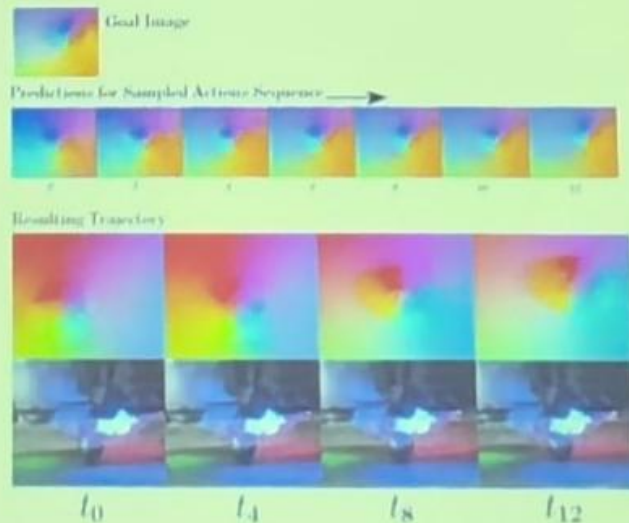
CVPR 2019, MIT



“触”到“视”跨
模态感知

动作 → 抓取 → 触感

ICRA 2019, UC Berkeley



从触觉中学习固定轨迹的动作

触感 → 生成 → 动作

ICRA 2019, MIT



触觉感知动作

触感 → 测量 → 动作

触觉数据采集过程耗时耗力耗材，如何高效产生数据？

视触觉传感系列综述之：触觉数据生成

Tactile Data Generation and Applications Based on Visuo-tactile Sensors: A Review

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ABSTRACT

Tactile sensation is an essential sensory system in humans, providing abilities such as perception and tactile feedback. Due to multimodality information that integrates tactile sensation, humans exhibit remarkable environmental understanding and dexterous manipulation capabilities. Considering the importance of tactile sensing, researchers have developed tactile sensors such as capacitive and piezoresistive types. In recent years, visuo-tactile sensors have won the favor of the research community with abstract tactile information visualization. The Visuo-tactile sensor is an innovative optical sensor that supports image-based tactile information with high resolution compared to electronic tactile sensors, offering new approaches for tactile dataset collection within multimodal datasets. Nevertheless, owing to challenges such as wear resistance, collecting visuo-tactile data remains a high-cost, low-efficiency task, which limits the development of tactile information in multimodal datasets. With the development of the generation methods, visuo-tactile data collection with low efficiency hopes to be solved. Considering the unique contribution of tactile data to multimodal datasets, this review focuses on visuo-tactile data generation. The generation methods for visuo-tactile sensors are categorized into two categories based on simulation approaches: 1) physics-based and 2) learning-based. Additionally, from the perspective of visuo-tactile data, the review summarizes the cutting-edge applications of multimodal datasets incorporating tactile information. Based on this, the challenges and future directions for development are discussed. This review serves as a technical guide for researchers in the field and aims to promote the widespread development and application of multimodal datasets incorporating tactile information.

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Tactile Data Generation and Applications Based on Visuo-Tactile Sensors: A Review

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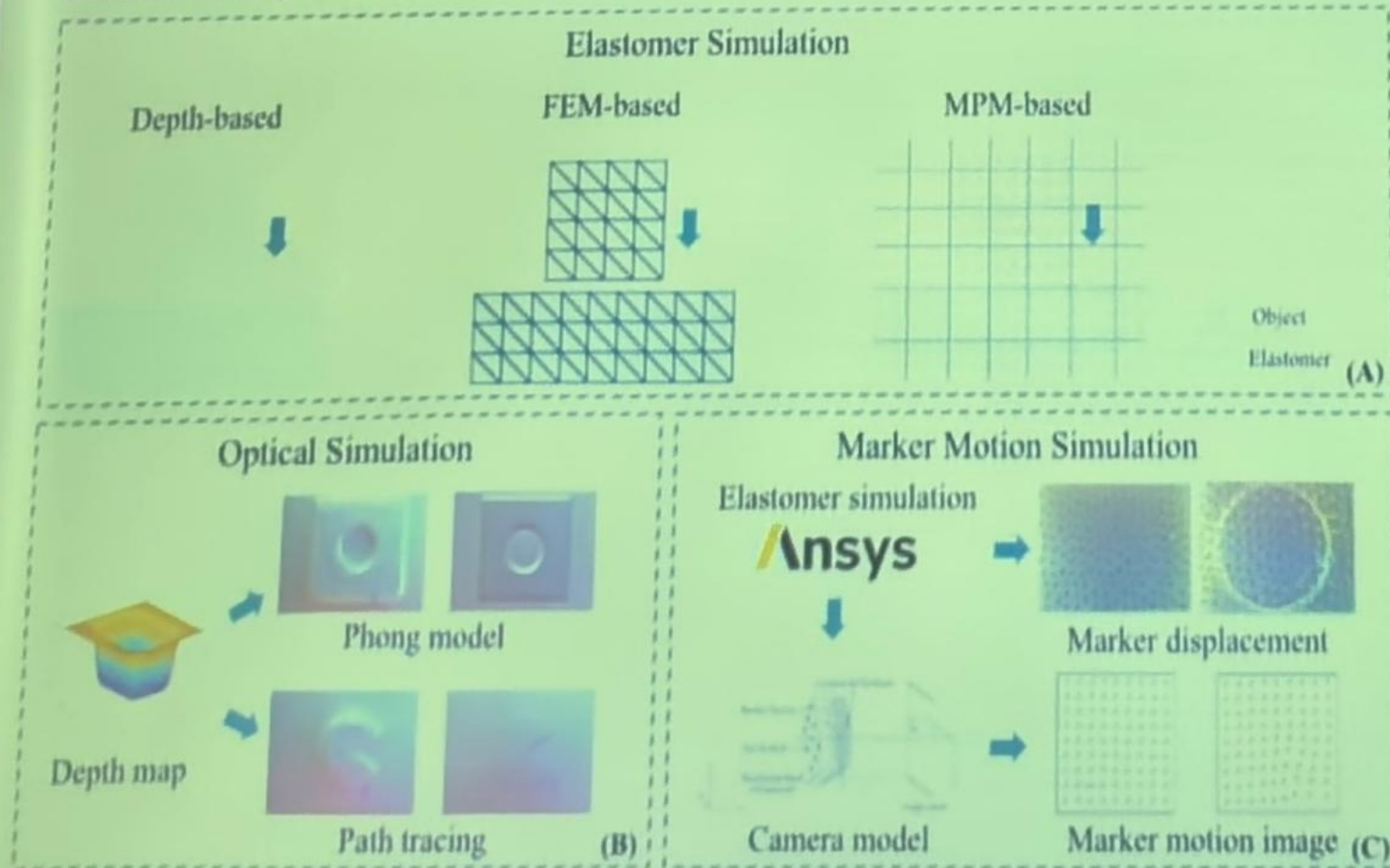
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The visuo-tactile data generation are divided into Elastomer simulation, Optical simulation, and Marker motion simulation



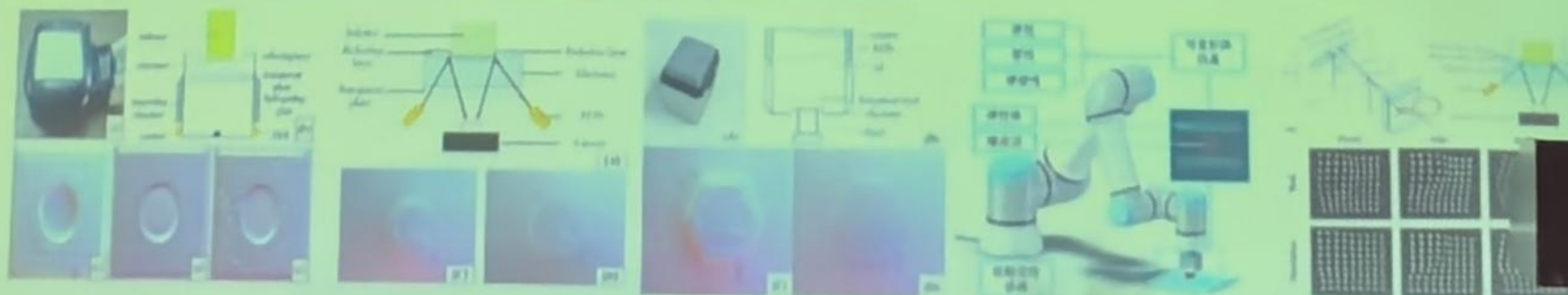
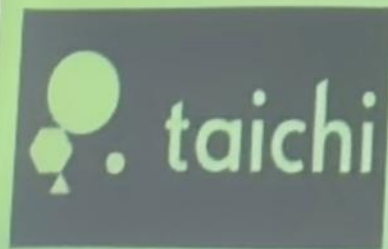
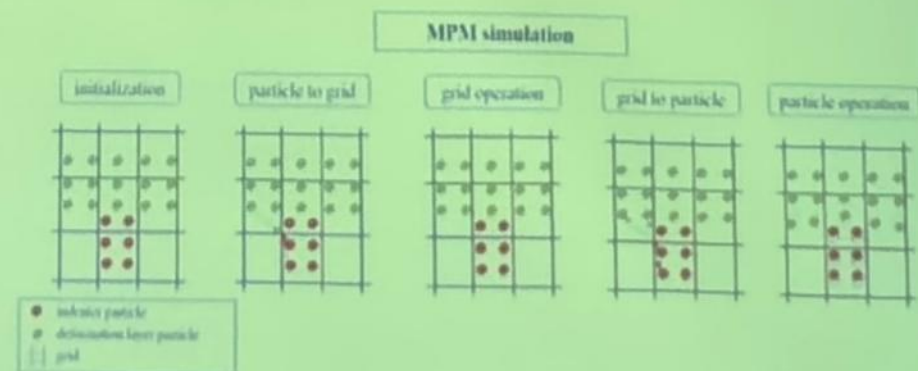
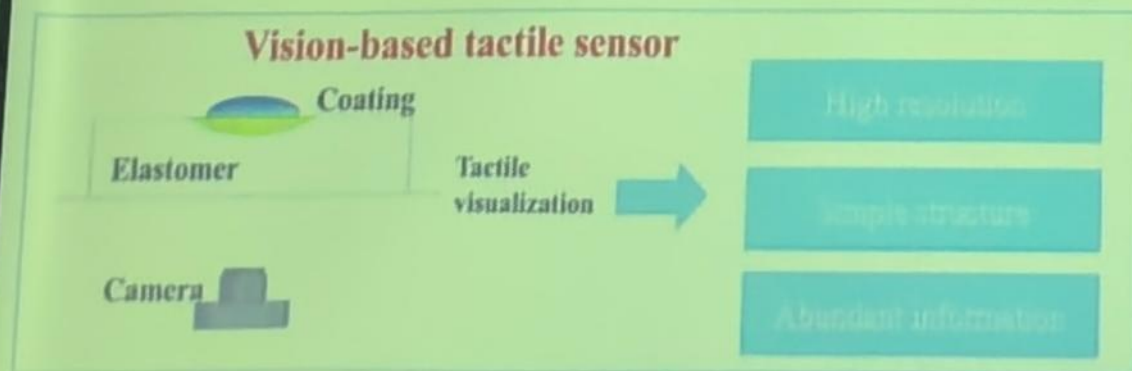
Elastomer Simulation: The simulation of the contact between object and elastomer.

Optical simulation: The simulation of camera and LEDs.

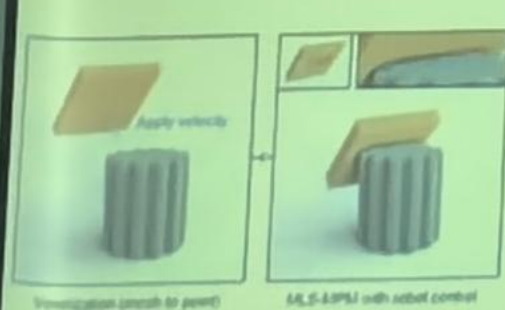
Marker motion simulation: The simulation of marker layer.

Tacchi 系列视触觉传感器接触仿真器

Tacchi 是利用移动最小二乘物质点法 (MLS-MPM) 和太极 (Taichi) 为基础完成的一系列视触觉传感器接触仿真器, 包括 Tacchi 1.0、Efficiency-tunable Tacchi、Multi-motion Tacchi、Soft-contact Tacchi 和 Tacchi 2.0。



Tacchi 系列视触觉传感器接触仿真器 - 发展历程



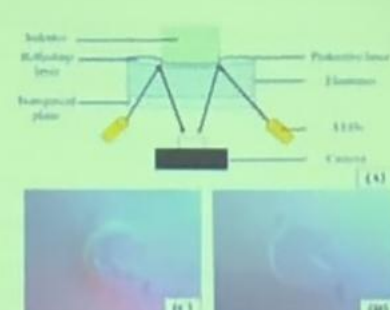
2021

Elastic tactile simulation



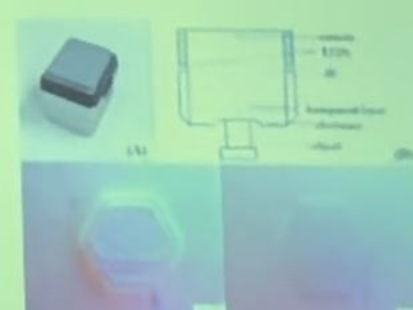
2023

Tacchi 1.0



2023

Efficiency-tunable Tacchi



2024

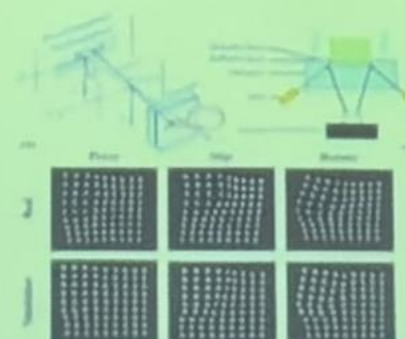
Multi-motion Tacchi



Future : TactileNet



To be continued



2024

Tacchi 2.0



2024

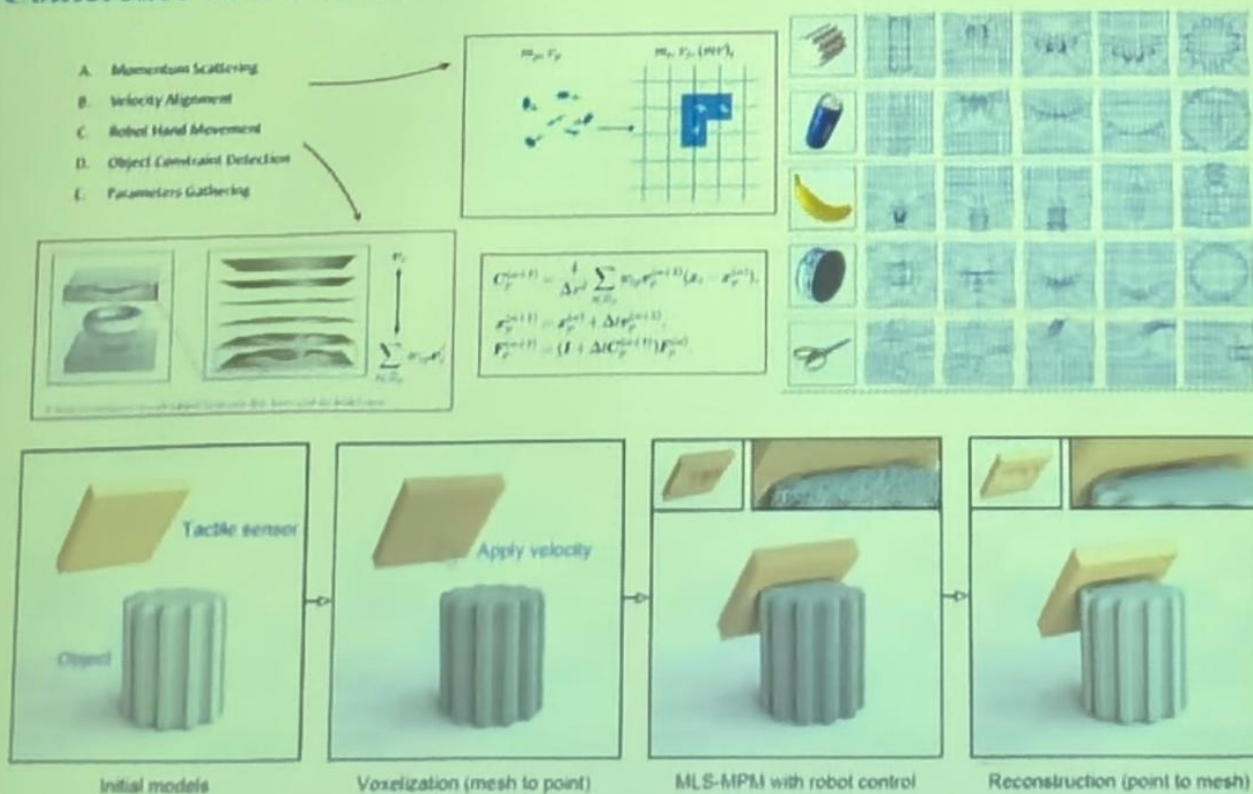
Soft-Contact Tacchi

CCF-A

Elastic Tactile Simulation Towards Tactile-Visual Perception

Anonymous ACM Multimedia Submission
Paper ID 1429

29th ACM International
Conference on Multimedia



Yikai Wang, Fuchun Sun*, Bin Fang. 29th ACM International Conference on Multimedia, pp. 2690 - 2698, 2021

Tacchi 系列视触觉传感器接触仿真器 - Tacchi 1.0



方法: MPM + Taichi + Phong model

能力: 触觉图像

接触物体: 刚体

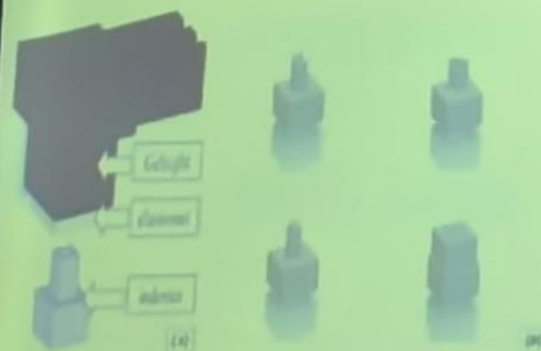
运动模式: 按压

优势: 低计算成本; 高鲁棒性和适用性; 可插拔。

TABLE I
IMAGE QUALITY COMPARISON

	SSIM $\uparrow$	PSNR $\uparrow$	MAE $\downarrow$
Tacchi_1	0.819 ± 0.011	18.43 ± 3.07	$8.73 \pm 0.03\%$
Tacchi_10	0.845 ± 0.008	18.57 ± 2.80	$8.57 \pm 0.03\%$
Tacchi_100	0.857 ± 0.000	18.46 ± 2.95	$8.67 \pm 0.03\%$
[1]'s method	0.852 ± 0.006	18.51 ± 3.26	$8.64 \pm 0.03\%$
[19]'s method	0.854 ± 0.005	18.50 ± 3.36	$8.66 \pm 0.03\%$

Tacchi 系列视触觉传感器接触仿真器 - Efficiency-tunable Tacchi



方法: MPM + Taichi + Path Tracing

能力: 触觉图像

接触物体: 刚体

运动模式: 按压

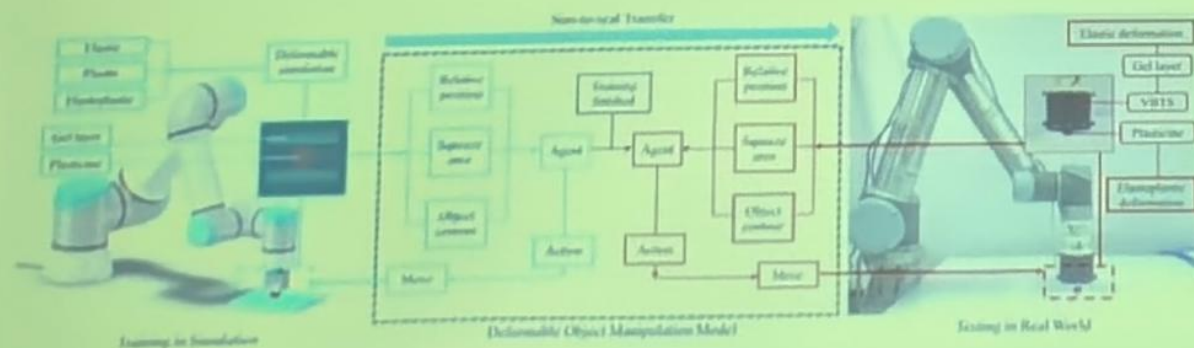
优势: 低计算成本; 高鲁棒性和适用性; 可插拔。

TABLE II
TIME USED

	Render_1 (480P)	Render_10 (320P)
moon	$21.585 \pm 2.111(s)$	$0.312 \pm 0.118(s)$
triangle	$12.837 \pm 1.831(s)$	$0.251 \pm 0.058(s)$
pacman	$17.977 \pm 2.563(s)$	$0.269 \pm 0.072(s)$
dots	$18.585 \pm 4.275(s)$	$0.273 \pm 0.157(s)$

[2] Sun Y, Zhang S, Shan J, and B. Fang*, et al. Simulation of Vision-based Tactile Sensors with Efficiency-tunable Rendering[C]//2023 IEEE International Conference on Robotics and Biomimetics (ROBIO). IEEE, 2023: 1-6.

Tacchi 系列视触觉传感器接触仿真器 - Soft-contact Tacchi



方法: MPM + Taichi + Von Mises Yield Criterion.

能力: 触觉图像

接触物体: 弹性、塑性、弹塑性

运动模式: 按压、滑移

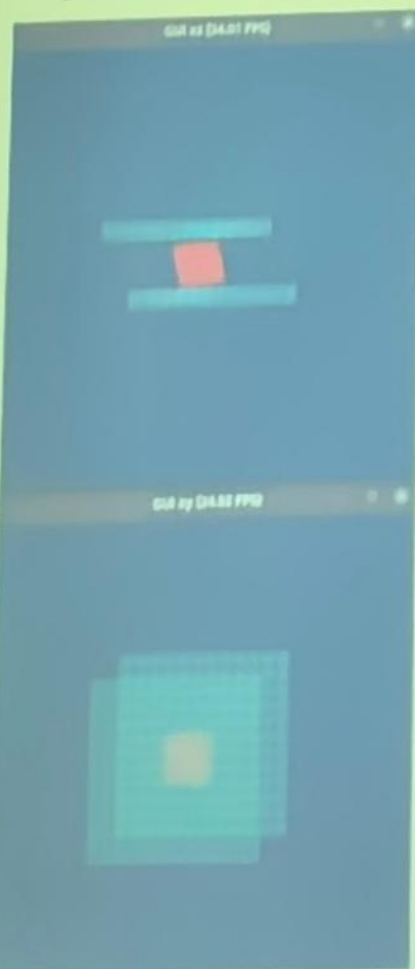
优势: 低计算成本; 高鲁棒性和适用性; 可插拔。

TABLE IV
AVERAGE ROUNDNESS AND THEIR STANDARD DERIVATIONS OF THE ROUNDNESS EVALUATION.

	Out of baseline†	Lack of satisfactory baseline‡
Cylinder (Front)	0.053	0.011
Sphere (Mean)	0.004	0.030

Tacchi 系列视触觉传感器接触仿真器 - Soft-contact Tacchi

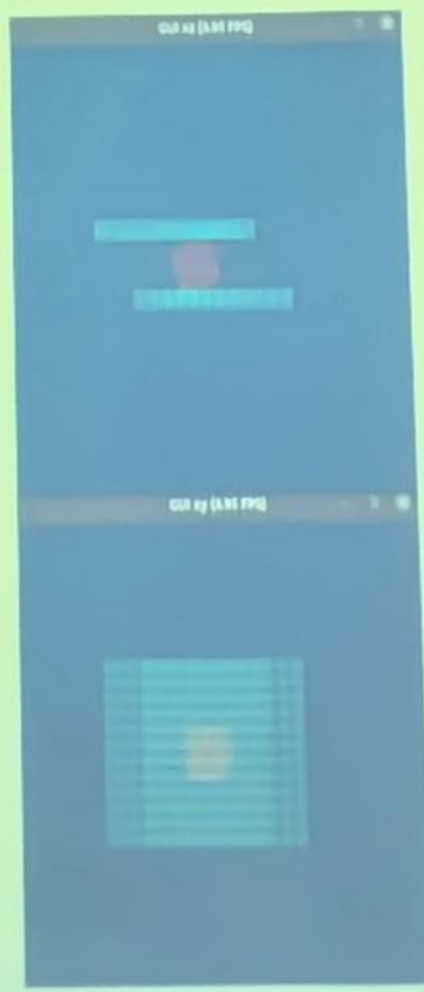
Position control



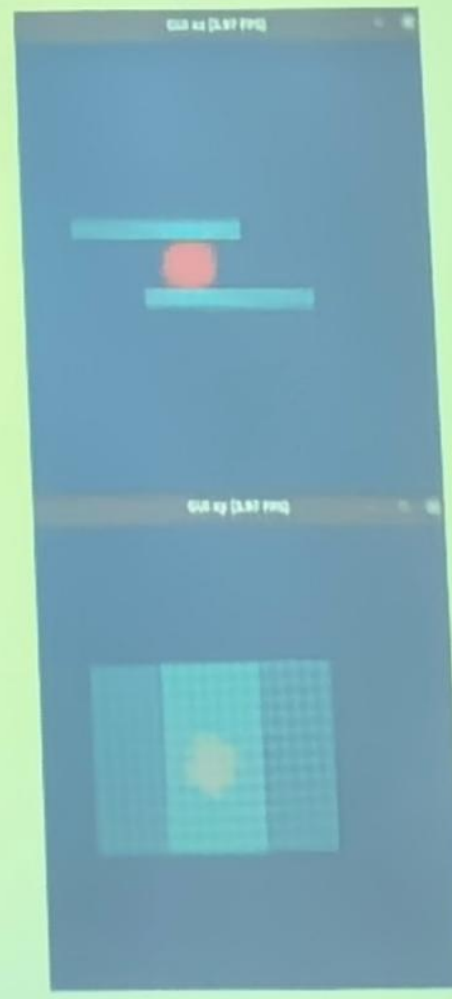
Squeeze



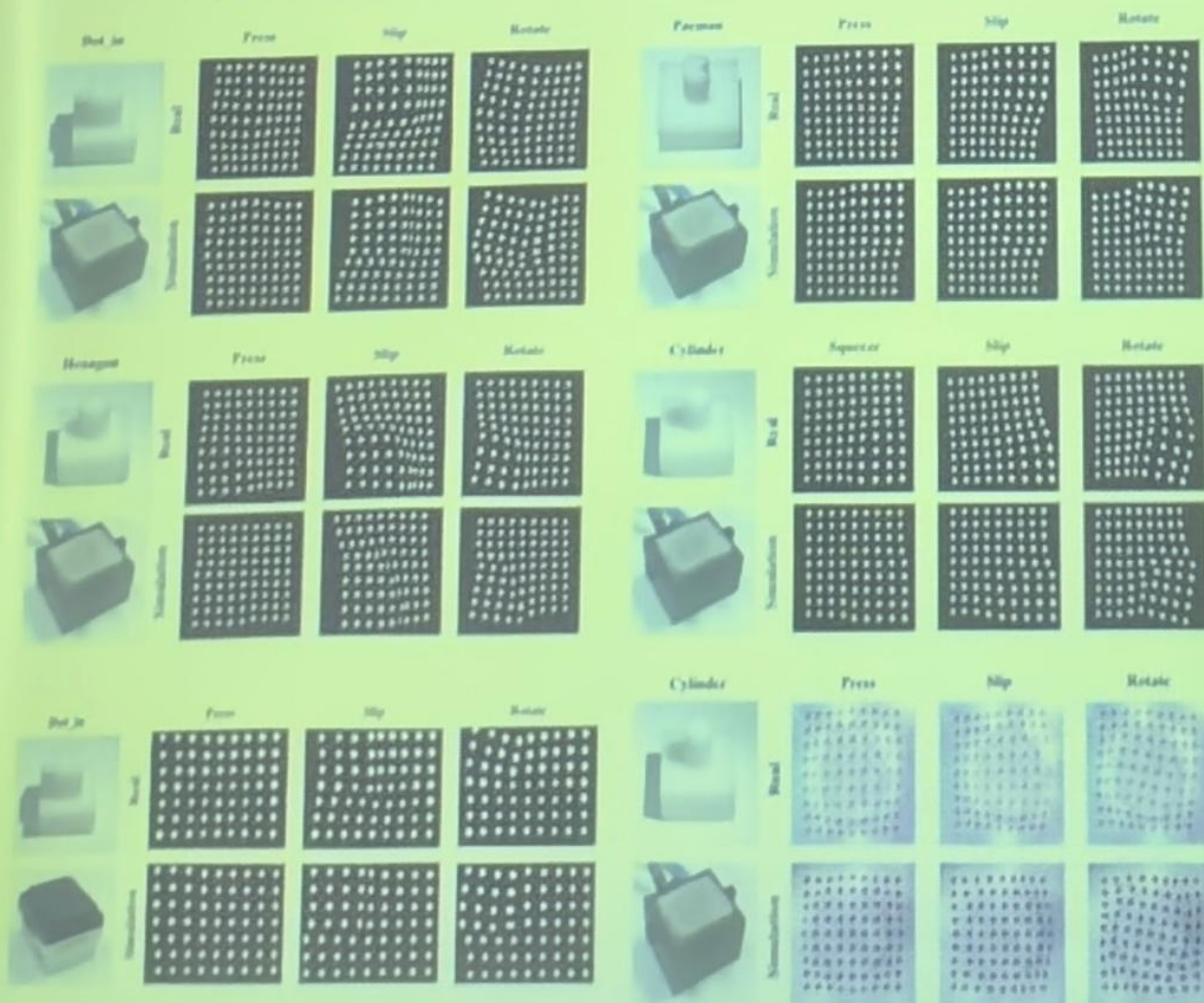
Cylinder



Sphere



Tacchi 系列视触觉传感器接触仿真器 – Tacchi 2.0



方法: IMPM + Taichi + Path Tracing + Camera Model

能力: 触觉图像、标记点图像、触觉标记点联合图像

接触物体: 刚体

运动模式: 按压、滑移、旋转、组合运动

优势: 低计算成本; 高鲁棒性和适用性; 可插拔。

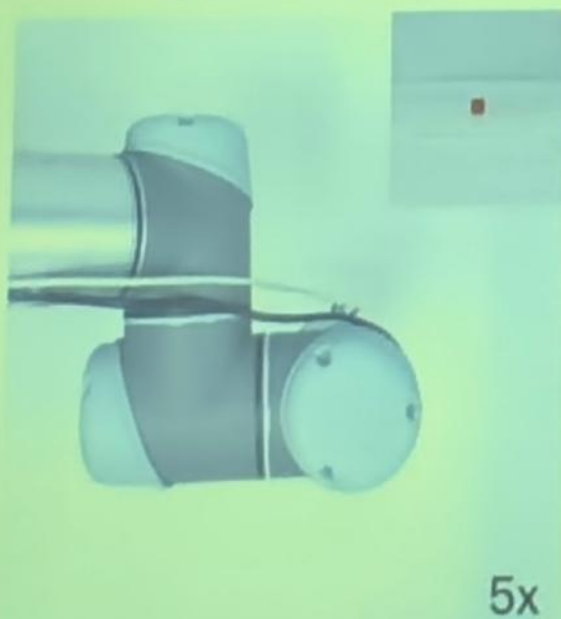
[5] Sun Y, Zhang S, and B. Fang*, et al. Tacchi 2 Computational Cost and Comprehensive Contact for Vision-based Tactile Sensors. (IROS2025审稿稿)

Tacchi 系列视觉触觉传感器接触仿真器 - Soft-contact Tacchi

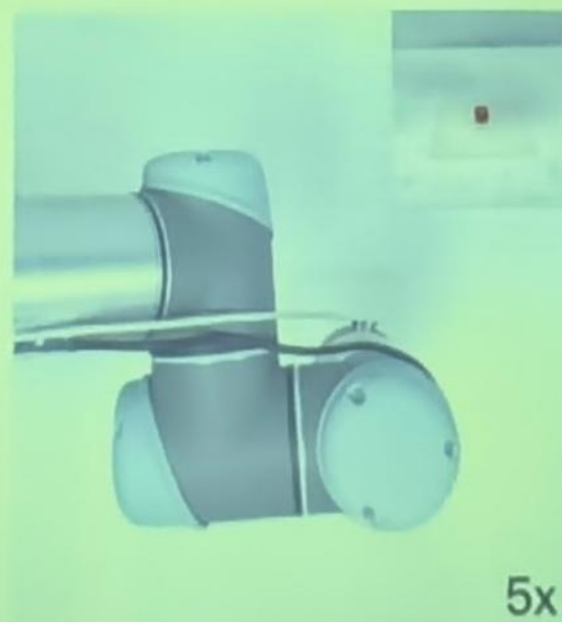
cylinder-pretraining



cylinder-multi



sphere-pretraining



sphere-multi

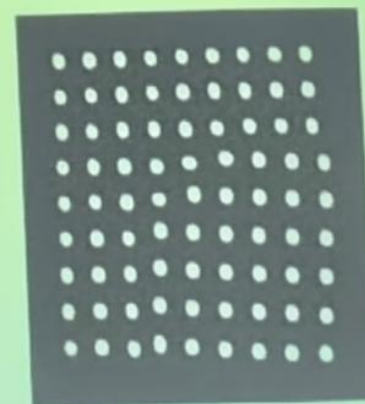
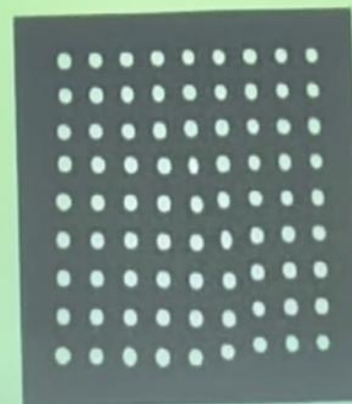
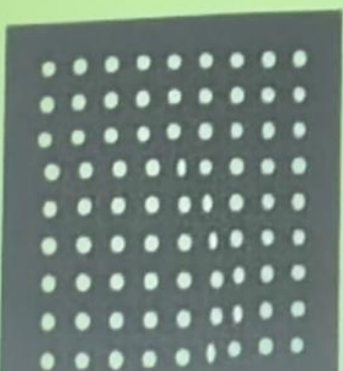
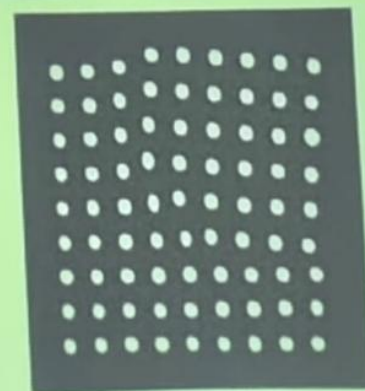
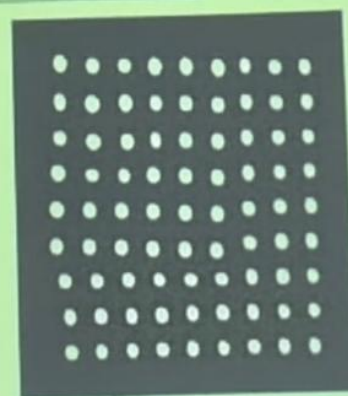
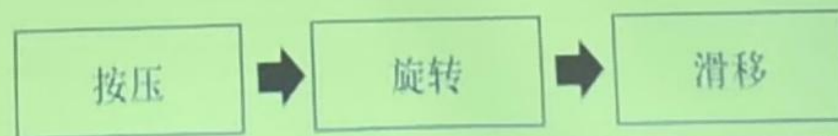
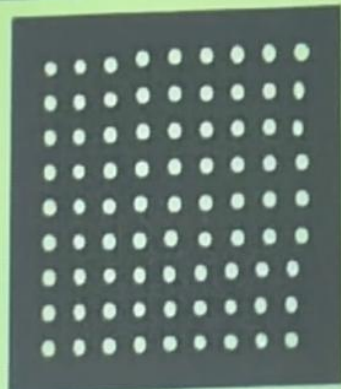
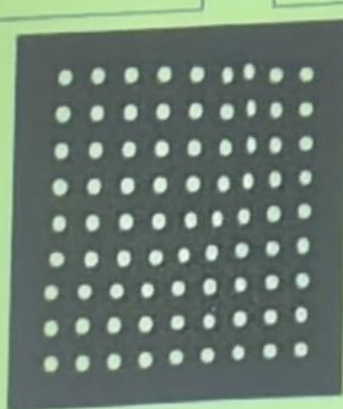
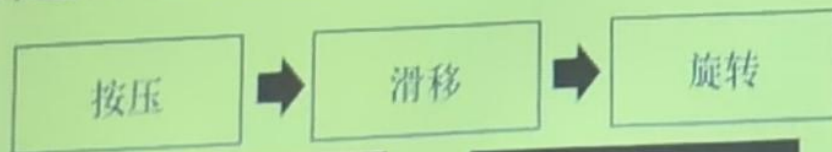


Sun Y, Zhang S, Shan J, **Fang B*** et al. Soft Contact Simulation and Manipulation Learning of Deformable Objects with

based Tactile Sensors [J]. arXiv preprint arXiv:2405.07237, 2024.

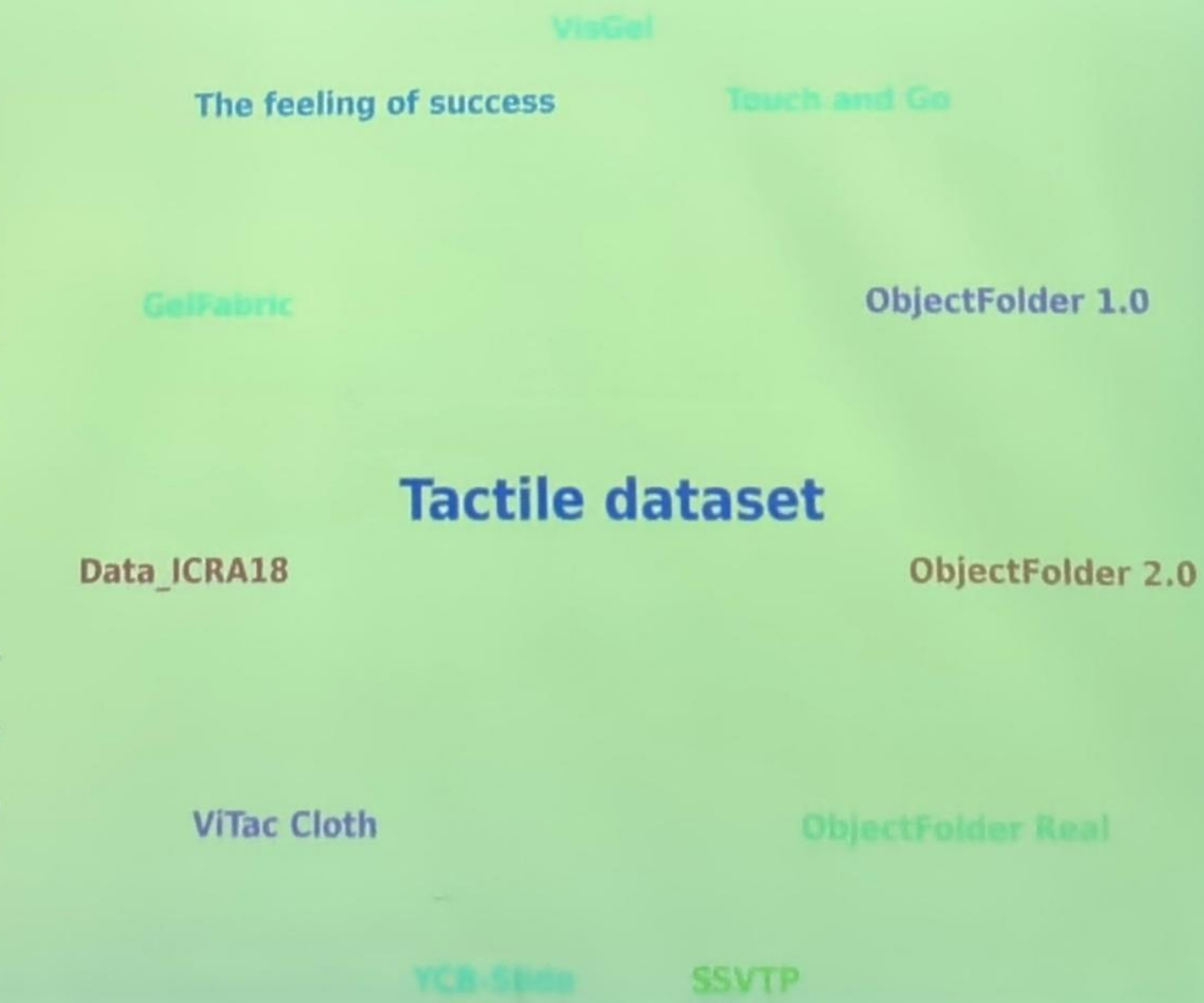
Tacchi 系列视触觉传感器接触仿真器 – Tacchi 2.0

标定点动态组合运动仿真:



触觉数据集

在自然环境中，信息以多种形式存在，单一模态的感知往往无法提供完整的理解。通过将触觉、视觉和语言等不同模态进行协同工作，机器人和AI系统可以获取更加全面和准确的环境信息。视觉为物体的形状、颜色和位置提供了直观的感知，触觉则补充了物体的物理特性，而语言可以传递关于这些特性的高层次语义描述。



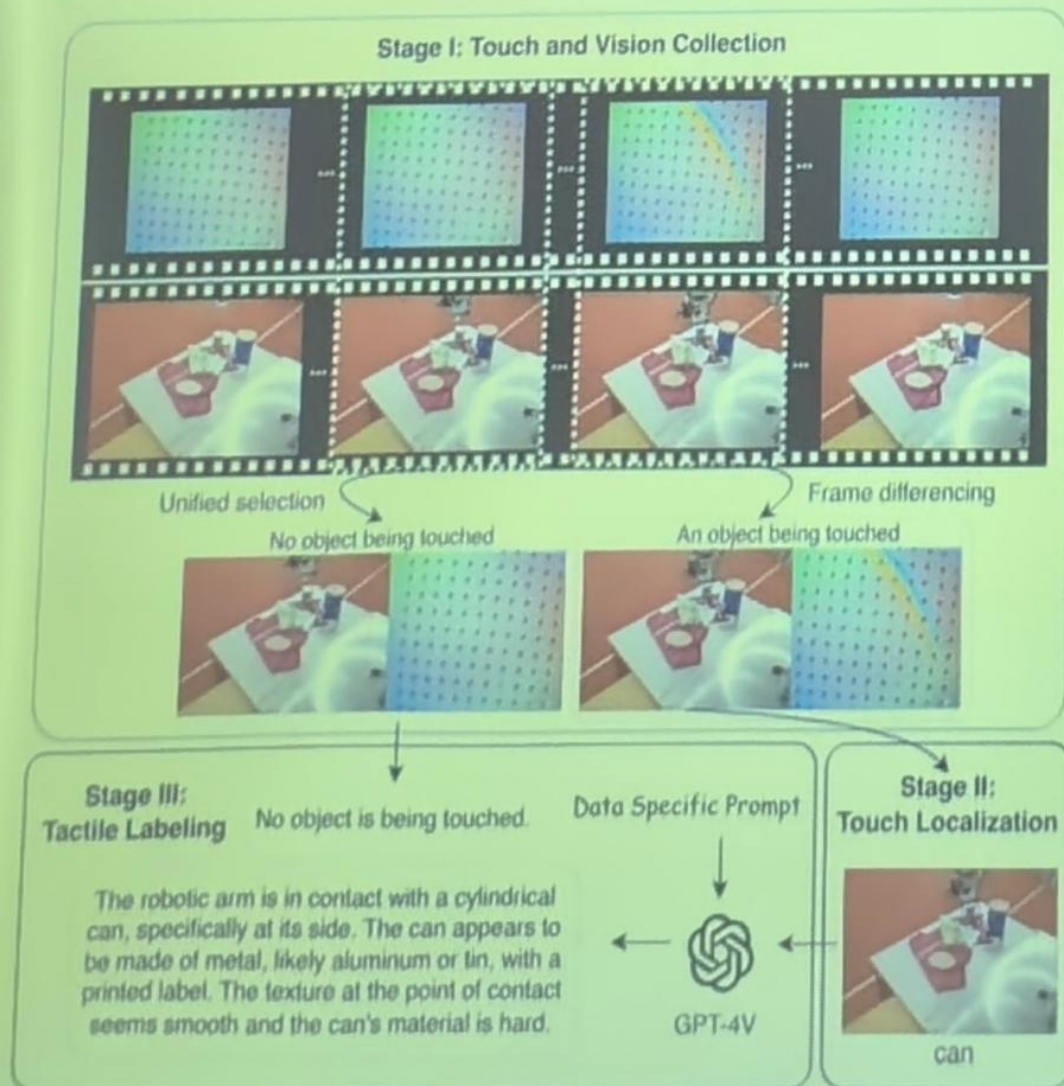
TLV数据集

TLV数据集是一个包含19,834条数据的触觉-语言-视觉数据集，其中语言是对触觉信息的句子级描述，该描述由人机级联协作注释。

构建过程分为三阶段：

1. 视触觉数据收集
2. 接触定位
3. 触觉信息标注

在触觉领域对语言模态的一个初步探索，并通过实验证明是有效的。



Touch100k数据集

Dataset	TM.	VM.	LM.	MG.	AS.	MO.	$\geq 100k$ DS.
Touch100k	✓	✓	✓	✓	✓	✓	✓
TLV [5]	✓	✓	✓	✗	✓	✓	✗
TVL [12]	✓	✓	✓	✗	✓	✓	✗
OF Real [14]	✓	✓	✓	✗	✗	✓	✓
SSVTP [25]	✓	✓	✓	✗	✗	✗	✗
TAG [43]	✓	✓	✓	✗	✗	✓	✓
OF 2.0 [15]	✓	✓	✓	✗	✗	✓	✓
YCB-Slide [36]	✓	✓	✓	✗	✗	✓	✓
OF 1.0 [13]	✓	✓	✓	✗	✗	✓	✗
VisGel [28]	✓	✓	✗	✗	✗	✓	✓
ViTac Cloth [31]	✓	✓	✗	✗	✗	✗	✗
Data_ICRA18 [46]	✓	✓	✓	✗	✗	✗	✗
GelFabric [47]	✓	✓	✗	✗	✗	✗	✗
Feel [4]	✓	✓	✓	✗	✗	✓	✗

项目主页:

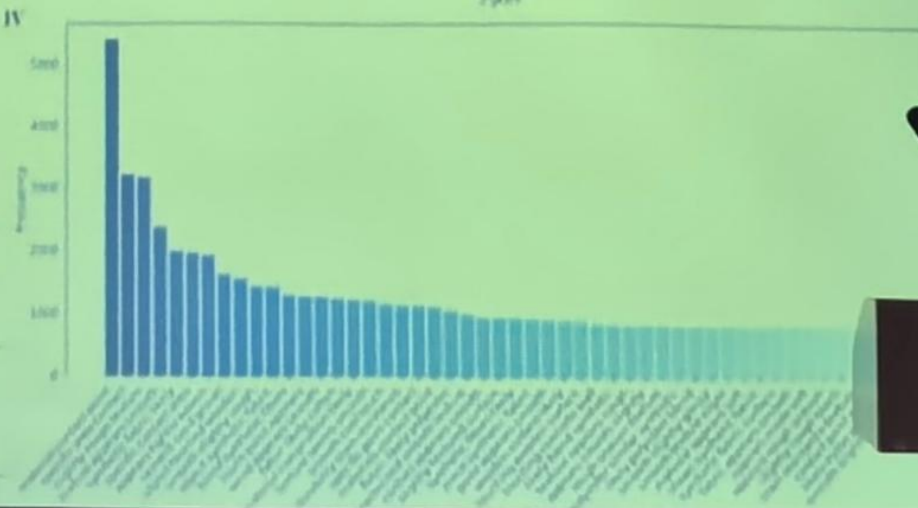
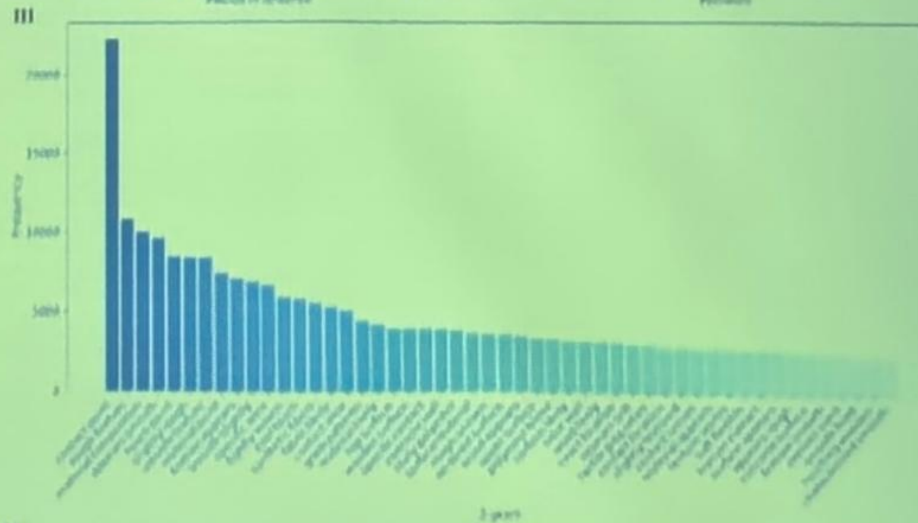
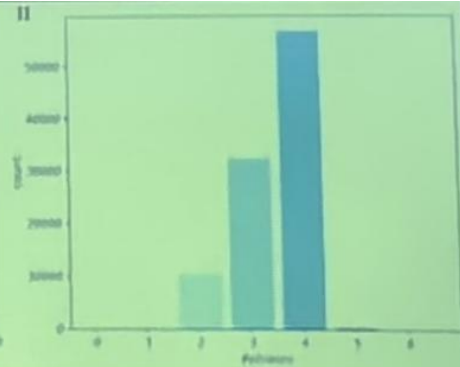
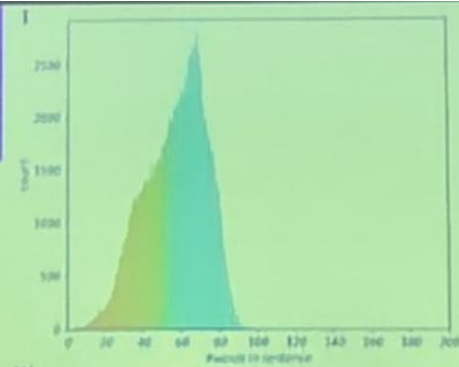
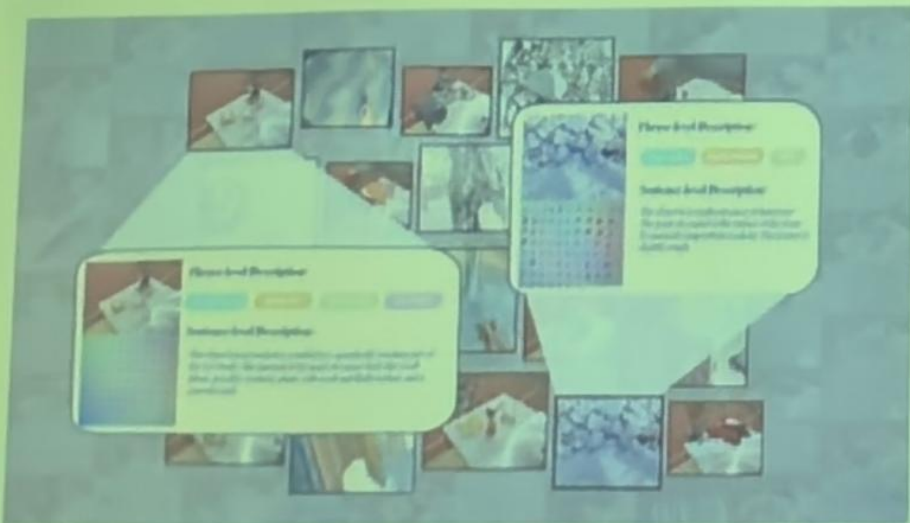
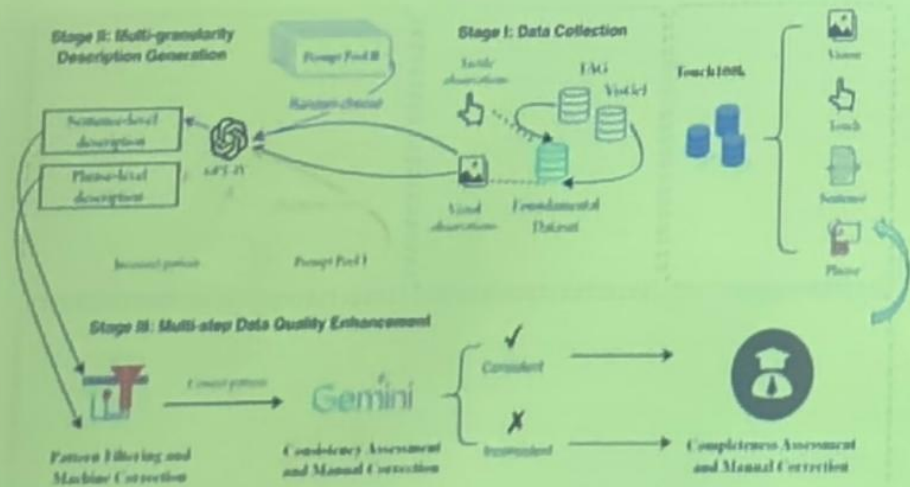
<https://cocacola-lab.github.io/Touch100k/>

Comparison of the Touch100k dataset to other tactile datasets.

TM.: Tactile Modality. **VM.:** Visual Modality. **LM.:** Language Modality. **ML.:** Multigranularity Language. **AS.:** Abundant Semantics. **MO.:** Multi-category Objects. **DS.:** Data Size.

Touch100k: A large-scale touch-language-vision dataset for touch-centric multimodal representation, Information Fusion,

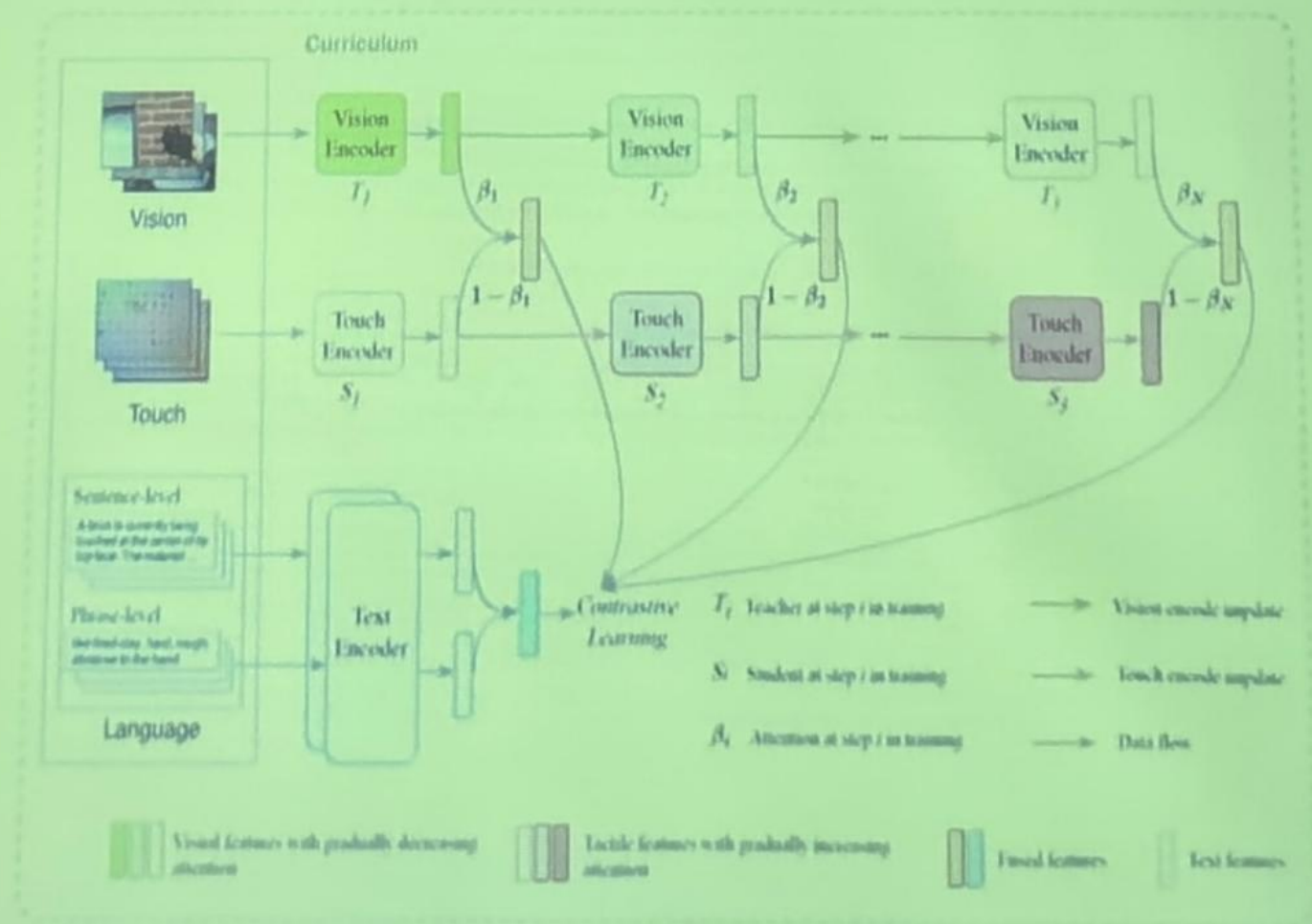
Touch100k数据集



方法

TLV-Link

- 用于触觉表示的课程学习



实验

Table 2: Accuracy of different models on *material property identification* and *robot grasping prediction* tasks. The best performance is **bold**, and the suboptimal is underlined. The best with performance improvement compared to the suboptimal within 0.5% are marked as green and above 0.5% are highlighted in red.

Model	Training Data	Material Property			Robot Grasping
		Material	Hard/Soft	Rough/Smooth	
Chance	-	5.0	50.0	50.0	50.0
<i>Linear Probing</i>					
Supervised	ImageNet	46.9	72.3	76.3	73.0
VT CMC [43]	TAG	54.7	77.3	79.4	78.1
MViTac [7]	TAG / Feel	57.6	86.2	82.1	-
UniTouch [42]	TAG, Feel, YCB-Slide, OF 2.0	61.3	-	-	<u>82.3</u>
VIT-LENS-2 [27]	TAG	<u>63.0</u>	<u>92.0</u>	85.1 (+0.4)	-
TLV-Link (Ours)	Touch100k	67.2 (+4.2)	93.1 (+1.1)	<u>84.7</u>	94.5 (+12.2)
<i>Zero-Shot</i>					
UniTouch [42]	TAG, Feel, YCB-Slide, OF 2.0	52.7	-	-	65.5 (+0.1)
VIT-LENS-2 [27]	TAG	<u>65.8</u>	<u>74.7</u>	<u>63.8</u>	-
TLV-Link (Ours)	Touch100k	70.0 (+4.2)	79.3 (+4.6)	77.8 (+14.0)	<u>65.4</u>



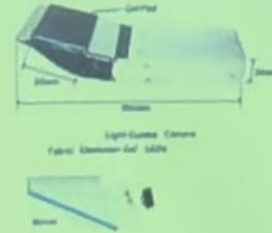
GelSight Mini



DIGIT



GelSight



GelSlim



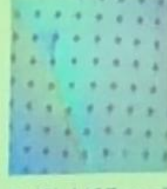
GelSight var.1



GelSight17 var.1



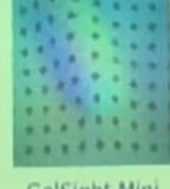
GelSight17 var.2



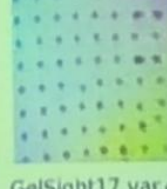
GelSight17 var.3



DIGIT



GelSight Mini



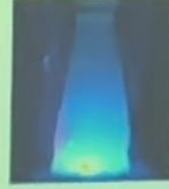
GelSight17 var.4



GelSight Wedge



GelSight 360



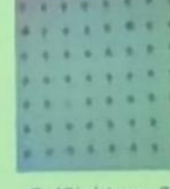
GelSight Finray



GelSight Svelte



DenseTact2.0



GelSight var.2

Low Standardization of Visuo-tactile Sensors^[7]

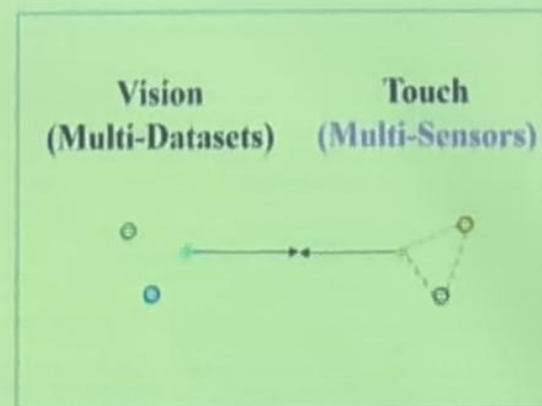
[7] J. Zhao, Y. Ma, L. Wang, and E. H. Adelson, "Transferable tactile transformers for representation learning across diverse sensors and tasks," *arXiv preprint arXiv:2406.13640*, 2024.

Motivation

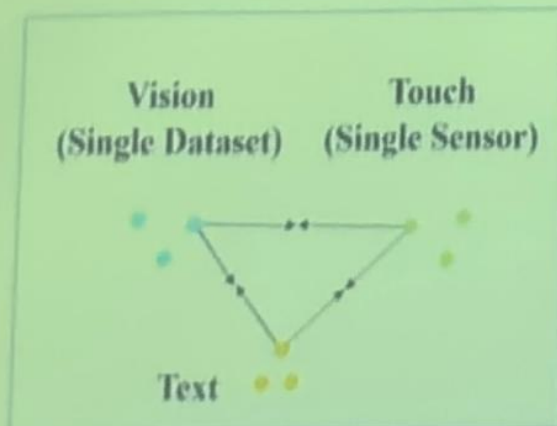
Touch-Vision Contrastive Learning



UniTouch



Touch-Vision-Language Contrastive Learning

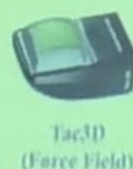


Representations with the same tactile information should be clustered!

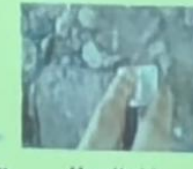
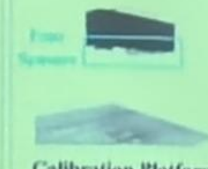
TacQuad Dataset

TacQuad: Aligned Multi-Modal **Multi-Sensor** Tactile Dataset

Four Visuo-tactile Sensors



Data Collection



Fine-grained Spatio-temporal Aligned



The tactile sensor touches the top center of a glass jar, which has a hard, smooth, and transparent surface with a series of texture.

25 Objects
30 Touches
17,524 Frames

Coarse-grained Spatial Aligned



The tactile sensor touches the top center of a rough hard orange. The contact point is relatively hard and firm with roughness and a slightly elastic feel.

(Indoor + Outdoor)
99 Objects
151 Touches
55,082 Frames

Material Distribution



Enabling Explicit Sensor-Agnostic Feature Extracting

AnyTouch 1.0

Multi-Level

Pixel-Level
Details



Semantic-Level
Features



Material
Roughness
Hardness
...

Static Sensing



Multi-Perspective

Dynamic Sensing



Multi-Sensor

Sensor Aligned
Data



Internet
Datasets

ObjectFolder

TAG



Experiments

- Sensor Transferability Analysis

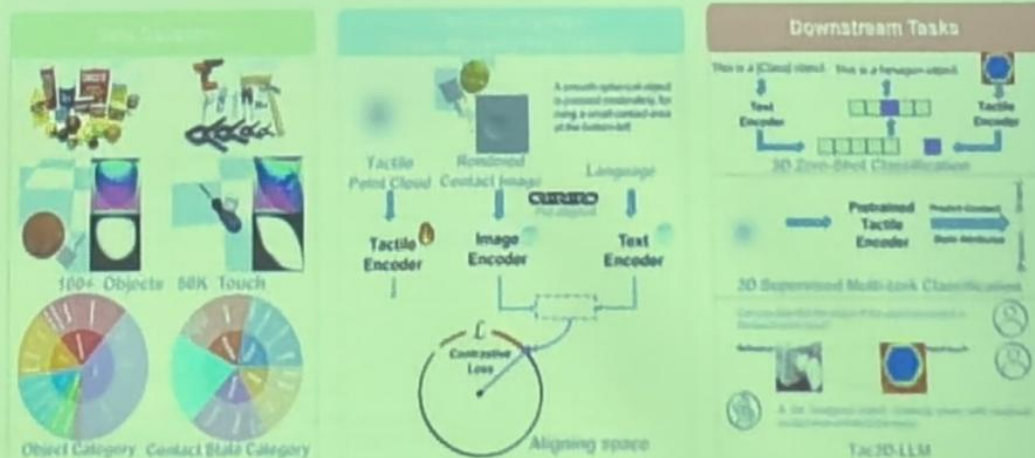
Tactile Training Data	Data Volume	TAG Material	Feel Grasp	OF 1.0 Material	OF 2.0 Material
No Tactile Pre-training (CLIP)	/	52.96	72.37	41.00	73.16
TAG, VisGel, Cloth	996k	83.55 (↑30.59)	79.12 (↑6.75)	46.12 (↑5.12)	75.10 (↑1.94)
TAG, VisGel, Cloth, OF Real	2161k	79.67 (↓3.88)	79.28 (↑0.16)	47.55 (↑1.43)	75.53 (↑0.43)
TAG, VisGel, Cloth, OF Real, TVL, SSVTP, YCB-Slide	2388k	79.61 (↓0.06)	79.10 (↓0.18)	48.00 (↑0.45)	75.57 (↑0.04)
TAG, VisGel, Cloth, OF Real, TVL, SSVTP, YCB-Slide, Octopi	2427k	79.70 (↑0.09)	79.40 (↑0.30)	48.75 (↑0.75)	75.66 (↑0.09)

介绍

本文提出了CLTP框架，通过构建数据集和预训练模型，实现触觉 3D 点云与自然语言的对齐，在多个下游任务中表现出色，为机器人触觉语言理解和操作提供了新方法。本文涵盖以下方面：

1. 数据集采集
2. CLTP训练
3. 下游任务验证

CLTP: Contrastive Language-Tactile Pre-training for 3D Contact Geometry Understanding, CoRL 2025



数据集

作者构建了一个数据集，其中包含了 YCB Objects, 自制的钉子和 McMaster 一共117个物件，基本涵盖生活和工业场景。同时，将数据集分为大小物体两组。对于每条采集的数据，记录(触觉点云，文字描述，触觉图像)三元组。

数据采集

模拟数据：使用作者先前工作TACTO触觉模拟器，根据不同策略在PyBullet中采集数据。

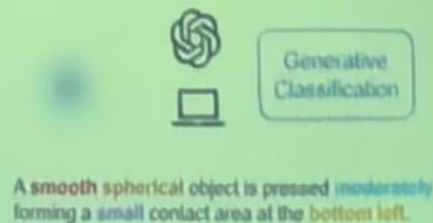
真实数据：通过GelStereo传感器采集点云数据，并且通过GelSight采集触觉图片



Contact State Definition

- Contact Shape (19): "spherical", "cylindrical", etc.
- Contact Area (5): "tiny", "small", "medium", etc.
- Contact Position (9): "top left", "top center", etc.
- Contact Force (4): "slightly", "moderately", etc.
- Texture (5): "smooth", "rough", "bumpy", etc.

Language Description Generation



机器人通用操作---**VTLA**，视觉-触觉-语言-动作大模型

虚拟环境仿真^[1]

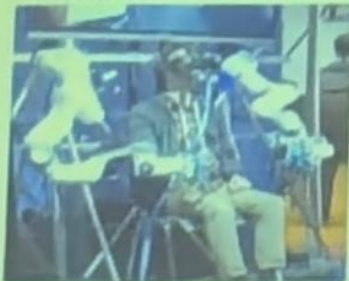
- 训练成本低
- Sim2Real Gap
- 英伟达



- Franka Kitchen
- Meta World

遥操作^[2]

- 直接有效
- 成本高
- 特斯拉



- RobotNet
- RoboTurk Real Robot
- Google Brain
- RoboSet

模仿学习^[3]

- 直接迁移人动作
- 可靠性无法保证
- 斯坦福 CMU



- Ego4D

动作库^[4]

- 标准化流程效率高
- 成功率低
- 谷歌



- RT1

- 通用触感灵巧手?
- 高质量多模态的操作数据?
- 通用操作学习能力?
- 内容生成技术用于操作?

欢迎交流合作!

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DeepTouch-Lab <https://deeptouch-lab.github.io/>

谢谢！